

SAVING MOTHERS & BABIES FOR THE NEW WORLD

Virtual Congress & Workshops:

25th to 28th August 2022

Hands-On Workshops:

**ADVANCED TWINS ULTRASOUND
WORKSHOP**

24th August 2022

**REGIONAL ADVANCED NEONATAL
AIRWAY AND VENTILATION WORKSHOP**

24th & 25th September 2022



21ST CONGRESS OF THE FEDERATION OF ASIA AND OCEANIA PERINATAL SOCIETIES (FAOPS) & 28TH CONGRESS OF THE PERINATAL SOCIETY OF MALAYSIA (PSM)

ISBN: 978-2-88971-033-1

DOI: 10.3389/978-2-88971-033-1

Citation: Federation of Asia and Oceania Perinatal Societies (FAOPS), Perinatal Society of Malaysia (PSM). (2022). 21st Congress of the Federation of Asia and Oceania Perinatal Societies (FAOPS) & 28th Congress of the Perinatal Society of Malaysia (PSM).



September 25–28, 2022, Virtual Congress & Workshops

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Welcome to the 21st Congress of the Federation of Asia and Oceania Perinatal Societies (FAOPS) & 28th Congress of the Perinatal Society of Malaysia (PSM)

The ambiguity of the new world may generate fear of the consequences of our present actions. The pride of our past accomplishments could have merely seemed to be of an avarice, exhausted due to the surge of the ubiquitous COVID-19 virus, the stalling economic state, and persistent wars. Despite the initiatives of the United Nation's sustainable developmental goal targets crafted during the pre-COVID era, the future of the world looks rather bleak with the health systems of many countries being further compromised. The hope lies in our new generation, which therein lies the need to save mothers and babies during this extraordinary time, and improve their outcomes in order to sustain the generation for the new world.

Good outcomes for mothers and babies can only be achieved by ensuring that health care professionals are knowledgeable, holistic, ethical, and safe; by ensuring continued acquisition of sound scientific knowledge, effective execution of ethically robust healthcare policies and guidelines, and by assuring equitable access to healthcare. These aims need to remain at the forefront of the policies of governments across all countries despite the present setbacks. The high healthcare burden to provide these, and the uncertainty of the future which gripped across Asia-Oceania countries, had therefore propelled the scientific committee to place the theme of "Saving Mothers and Babies for the New World" as the aim of the FAOPS 2022 congress.

This collection comprises a total of 80 abstracts that were submitted and presented at the congress that was successfully convened as scheduled.

LIST OF ORGANIZERS

Federation of Asia and Oceania Perinatal Societies (FAOPS)

Perinatal Society of Malaysia (PSM)

Abstract ID: A-0006

E-Poster

A case report: Maple syrup urine disease encephalopathy

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Maple syrup urine disease (MSUD) is a rare metabolic condition that affects one in 185,000 births of the population, irrespective of gender. It is passed down genetically by autosomal recessive inheritance. The name of the disease takes after the smell of the urine when the serum amino acid, isoleucine, is raised as a result of the defect in amino acid degradation pathway. An increase in branched-chain amino acids such as leucine, isoleucine and valine can cause multisystem disorders if it is left undetected.

We report a case of a newborn male with underlying Glucose-6-Phosphate Dehydrogenase (G6PD) deficiency, who had presented at the 3rd week of life with weight loss, poor feeding and was less active. He was hypotonic and had bradypnoea with unequal sluggish pupils. The reflexes were brisk in his limbs and the anterior fontanelle was tensed and bulging. Maple syrup odor was present in the urine. The blood results showed metabolic acidosis while radio-imaging of the brain revealed cerebral edema. Electroencephalogram (EEG) recording showed the absence of brain activity. Marked elevation in the leucine, isoleucine, valine and allo-isoleucine in the inborn errors of metabolism (IEM) screening was highly suggestive of maple syrup urine disease.

Although MSUD is a rare incidence, it is essential to rule out metabolic cause if the patient is not thriving well. MSUD may be fatal if not treated early, as high branched chain amino acids will accumulate in the body leading to progressive encephalopathy. Early intervention by BCAA-restriction diet and close follow up on nutritional and growth parameters are the mainstay of treatment.

Keywords: Maple syrup urine disease, encephalopathy, inborn errors of metabolism, branched-chain amino acids

Abstract ID: A-0007

E-Poster

Non-Immune hydrops fetalis: Case series from Sabah Women and Children Hospital

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INTRODUCTION

Hydrops fetalis is a clinical condition characterized by the pathological fluid accumulation in fetal serous cavities, involving at least 2 compartments. Skin edema is often associated with this condition. The basis of the disorder is an imbalance in the regulation of fetal fluid movement between the vascular and interstitial space. Hydrops fetalis is broadly classified into Immune Hydrops Fetalis (IHF) and Non-Immune Hydrops Fetalis (NIHF). The widespread use of anti-D immunoglobulin has resulted in a decline in the incidence of iso-immune fetal hydrops, but an increasing prominence of non-immune causes of this severe and highly lethal condition.

REPORT

We hereby report 4 cases of non-immune hydrops fetalis encountered at Sabah Women and Children Hospital (SWACH) in the year of 2021, looking into the different causes for this condition. They were all associated with preterm delivery ranging 30 weeks to 34 weeks. One out of 4 of them were detected antenatally. Out of these four cases the mortality rate was 75%. The average length of stay for the expired cases was around 1 day. One of the cases was discharged home after 85 days of stay in our hospital. The causes of these 4 cases of NIHF were placental chorioangioma, twin-to-twin transfusion syndrome, trisomy 21 and cardiomyopathy.

CONCLUSION

The mortality of non-immune hydrops fetalis is still high despite recent advances. It is important to detect and determine the cause of the non-immune hydrops fetalis in order to administer optimal multidisciplinary management during the perinatal period. In view of the high morbidity and mortality rate, parental counselling should be initiated early.

Keywords: Placental chorioangioma, Twin-to-twin transfusion syndrome, Trisomy 21, Cardiomyopathy

Abstract ID: A-0008

E-Poster

Thyrotoxicosis during pregnancy diagnosed from thrombocytopenia

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INTRODUCTION

Thyrotoxicosis is a severe condition with the potential to develop into a thyroid crisis, which can be triggered by delivery. However, because universal screening for thyroid dysfunction has not been recommended, it is very difficult to diagnose in asymptomatic pregnant women. Although thrombocytopenia has been linked to hyperthyroidism, there have been few reports of pregnant cases. Here we report a successfully managed thyrotoxicosis case during pregnancy diagnosed from thrombocytopenia detected in routine prenatal visit.

REPORT

A Japanese primiparous pregnant woman with well-controlled gestational diabetes mellitus (GDM) in her 35th week gestation was referred to our hospital due to progressing thrombocytopenia with a platelet count of $7.4 \times 10^4/\mu\text{L}$. Screening for autoimmune diseases revealed her thyrotoxicosis (TSH 0.008 $\mu\text{IU/mL}$, FT4 3.36 ng/dL, FT3 7.12 pg/mL, negative for thyroid autoantibodies). Ultrasonography detected a thyroid nodule of 20 mm and hyperperfusion around it, which suggested she was suffering from Graves' disease or Plummer's disease. Her thrombocytopenia seemed autoimmune. Fetal ultrasonography found her fetus to be normal. Multidisciplinary discussion concluded that her condition was at high risk for thyroid crisis at delivery. Antithyroid drug therapy (methimazole and iodine) was started, which suc-

cessfully stabilized her condition. In the 39th week, a non-stress test recorded a prolonged deceleration, and after failed labor induction she underwent an emergent C-section. She delivered a healthy baby weighing 3,225 g with Apgar scores of 8 and 10 at 1 and 5 minutes, respectively, and normal thyroid function. Her postpartum course was uneventful, and her thyroid function was stable with methimazole at 2 months postpartum. In conclusion, thyroid function tests should be considered in thrombocytopenic pregnant women.

Keywords: pregnancy, thyrotoxicosis, hyperthyroidism, thrombocytopenia

Abstract ID: A-0009**Oral****A 2-year analysis on outcome of infants of diabetic mothers (ill babies) admitted to NICU Taiping in 2019-2020****Arathi K^{1*}, Noor Zakirah N², Rohani AJ²**¹Department of Paediatrics, Neonatal Intensive Care Unit, Hospital Raja Permaisuri Bainun, Perak, Malaysia²Department of Paediatrics, Neonatal Intensive Care Unit, Taiping Hospital, Perak, Malaysia

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OBJECTIVE

To determine the outcome, mainly respiratory and cardiac complications of infants born to mothers with Gestational Diabetes Mellitus and pre-existing Diabetic, and to analyze the differences in outcome according to the type of maternal diabetes and their modes of treatment.

METHODS

It was a retrospective cohort study. Medical records of neonates who are infants of diabetic mothers (IDM) admitted to NICU for ventilatory support for the period of January 2019-December 2020 were reviewed.

BACKGROUND

Diabetes is the most common medical complication in pregnancy, affecting about 0.5-5% of all pregnancies. Infants of diabetic mothers are at increased risk of periconceptional, fetal, neonatal and long-term complications such as: diabetic embryopathy, hypoglycemia, hypocalcemia, polycythemia, myocardial hypertrophy and delayed lung maturation.

RESULTS

A total of 296 IDM babies were admitted to NICU Taiping Hospital from 2019 to 2020 for ventilatory support. Out of these 296 babies, data of 200 babies were analyzed. We observed a mortality rate of 9%. The commonest cause of death was persistent pulmonary hypertension (PPHN) which carried a 66.67% mortality rate. Highest percentage of this death is among the term babies with good birth weight. Term RDS complicated by PPHN contributes to this high rate of mortality in term babies. Infants born to mothers with GDM on diet control have poorer outcomes as compared to GDM on treatment (oral hypoglycaemic agent and Insulin). Significant association was found between various complications in the infants and maternal glycemic control (based on HbA1c levels).

CONCLUSION

Various complications are seen in IDMs, which consequently increases the morbidity and mortality in these babies. Strict glycaemic control in the mothers, planned pregnancy, proper antenatal care and strict monitoring in babies are required to prevent morbidity and mortality in IDMs.

Keywords: Gestational Diabetes Mellitus (GDM), Infant of Diabetic Mother (IDM), Persistent Pulmonary Hypertension in Newborn (PPHN), Respiratory Distress Syndrome (RDS)

Abstract ID: A-0010

E-Poster

TRPV4 mutation in congenital distal spinal muscular atrophy: A case report

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BACKGROUND

TRPV4 (transient receptor potential vanilloid 4 cation channel) mutation is associated with a spectrum of neuropathies and skeletal dysplasias inherited in an autosomal dominant pattern. It is expressed mainly in bones, skin and motor neurons. Patients affected with TRPV4 mutation can present with congenital or late onset of progressive neuropathy, skeletal abnormalities and respiratory failure. We herein describe a rare case of a neonate with TRPV4 mutation, who had suffered from respiratory failure and neurological disabilities at birth.

REPORT

We report a female neonate who was born at 37 weeks with a low birth-weight of 2.27kg, via an elective lower segment caesarean section for intra-uterine growth restriction and breech presentation. Antenatally, the mother had Grave's Disease and was on treatment. A detailed fetal scan done on the 36th week showed excessive scalp hair with lower limb deformities. The baby was born vigorous but was intubated at 15 min of life in view of bradycardia and poor saturation. At birth, the baby was hypotonic and areflexic, with no spontaneous limb movement or facial grimacing. She also had muscle atrophy and arthrogryposis multiplex. Bedside echocardiogram, ultrasound cranium and eye screening were normal. Her creatine kinase was high. There was no family history of neuromuscular or skeletal disorder. Her whole exome sequencing showed a pathogenic missense variant in TRPV4 gene (p.Arg269His). This variant is known to cause autosomal dominant distal

neuronopathy, also known as Congenital Distal Spinal Muscular Atrophy. Both parents refused for further family genetic testing. Parents wished for palliative management and withdrawal of care was done on day 11 of life.

CONCLUSION

Prevalence of the TRPV4 associated conditions has not been well studied in Malaysia. There is a wide spectrum of disorders associated with TRPV4 mutation, although overlap within each other can happen. Genetic studies are necessary to confirm the diagnosis. Management of this condition is mainly supportive and genetic counselling.

Keywords: TRPV4, Neuropathies, Skeletal Dysplasia, Genetics

Abstract ID: A-0011**E-Poster****Comparison of neonatal outcomes of small for gestational age and appropriate for gestational age preterm infants born at 34 to 36 weeks of gestation**

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BACKGROUND

Small for gestational age (SGA) are known to have higher perinatal morbidity and mortality as well as morbidity risk later in life as compared to appropriate gestational age (AGA) infants. The risk factors for SGA are complex and may be caused by maternal, placental or fetal factors.

OBJECTIVES

To study the risk factors and perinatal complications in late preterm SGA infants.

METHODS

This is a retrospective case control study of late preterm infants admitted over a 2-year period (January 2019- December 2020) in NICU Hospital Sultan Ismail Johor Bahru. The risk factors were analyzed and the frequencies of perinatal complications were compared between the two groups.

RESULTS

A total of 96 SGA and 229 AGA infants were analysed. The risks of SGA were significantly higher among mothers with pregnancy induced hypertension (aOR: 4.049; 95% CI: 2.263-7.244) and oligohydramnios (aOR 3.628; 95% CI:1.086-12.119). There were no significant differences among the perinatal complications studied but the SGA group achieved full feeding later (p value: 0.018) and had a longer hospital stay (p value: 0.031) compared to the AGA group.

CONCLUSION

In conclusion, SGA is the result of multifactorial interaction and can cause perinatal complications. By strengthening the perinatal care and monitoring, the complications of SGA can be reduced.

Keywords: SGA, perinatal outcome

Abstract ID: A-0012

E-Poster

Glucose monitoring among infants of diabetic mothers on treatment admitted to special care nursery in a tertiary center - A retrospective study

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INTRODUCTION

Neonatal hypoglycemia can lead to long-term neurodevelopmental impairment. Infants of diabetic mothers (IDM) are at high risk of developing hypoglycemia in the immediate hours after birth. Various international guidelines recommend screening of asymptomatic IDM for hypoglycemia.

OBJECTIVES

This study aims to determine the incidence of hypoglycemia and to review our local management among IDM on treatment admitted to Special Care Nursery (SCN) in a tertiary center – in order to conclude whether there should be changes implemented to our current practice.

MATERIAL AND METHOD

This was a single center, retrospective study involving healthy term and late preterm IDM on treatment who were admitted to SCN from 1st May 2021 to 30th September 2021 for blood glucose monitoring. These infants were identified from the admission registry; medical records were then traced and reviewed.

RESULT

Overall incidence of hypoglycemia from 476 infants was 0.6% ($N=3$). All the three infants' mothers were on combination therapy of insulin and oral hypoglycemic agent (OHA); only one of the three infants had symptomatic hypoglycemia.

The median age on admission and first feed given was 38 and 73 minutes of life respectively. Only 24.2% of infants were given their first feeding by one hour of life. First blood glucose was obtained at a mean age of 2 hours. The median number of blood glucose heel pricks done for each infant was 9 (range 3-39) and monitoring was stopped at a median age of 27 hours.

CONCLUSION

Hypoglycemia incidence was low among this cohort of infants. This needs to be confirmed by a properly designed observational or randomized controlled trial. A local protocol should be designed to standardize blood glucose monitoring with a potential to reduce the number of painful heel pricks.

Keywords: Hypoglycemia, Infant of diabetic mother

Abstract ID: A-0013**E-Poster****A case report: Pulmonary artery thrombosis in a newborn with severe dehydration****Chia Chia Tan, Pooven Raj Selva Rajah*, Rohani Abdul Jalil****Department of Paediatrics, Neonatal Intensive Care Unit, Taiping Hospital, Perak, Malaysia***raj.pooven1723@gmail.com; rohaniabduljalil@yahoo.com***BACKGROUND**

Pulmonary artery thrombosis in the newborn is a rare occurrence. It is approximately 1 in 40,000 births, with 90% of cases linked to indwelling intra-arterial catheters. Very few cases of spontaneous neonatal pulmonary arterial thrombosis have ever been described.

CASE REPORT

We report a 10-day old baby who was born at term gestation of 38 weeks, presenting to us with a history of watery stools and feeding intolerance. He was critically ill, having 16% weight loss with severe dehydration and was in hypovolemic shock. He required aggressive fluid resuscitation and inotropic support with the initial blood investigation showing acute kidney injury with severe metabolic acidosis. Bedside echocardiogram showed an incidental huge mobile intracardiac mass at the right ventricular outflow tract measuring 0.53 cm². Excluding the possibility of cardiac tumors, we initiated unfractionated heparin infusion, followed by low molecular weight heparin for a total treatment duration of 3 months. Serial echocardiogram during admission showed that there was reduction in the size of thrombus and final resolution of it was seen prior to discharge home at day 50 of heparin treatment.

CONCLUSION

Neonatal pulmonary artery thromboembolic events are infrequent. However, it can occur in neonates particularly with predisposing factors such as sepsis, dehydration, asphyxia, indwelling central lines, and prematurity. This case highlights the importance of not underestimating dehydration as a cause of thrombosis.

Keywords: Neonatal, Thrombosis

Abstract ID: A-0014

E-Poster

Neonatal tuberous sclerosis complex - A case report

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BACKGROUND

Tuberous sclerosis complex (TSC) is a rare genetic multisystem disorder causing non-cancerous tumours (hamartomas) in the brain, heart and other vital organs. Incidence of TSC is about 1 in 6000 livebirths. Cardiac rhabdomyomas, arrhythmias and cerebral lesions detected on antenatal scans are the major presenting findings in the fetus.

REPORT

Here we report a male infant born at term with a presumptive diagnosis of TSC. Fetal scan had demonstrated multiple huge rhabdomyomas and subependymal giant astrocytoma. Postnatal examination and imaging studies confirmed the antenatal scan findings. He did not have clinical seizures but his EEG showed sharp wave discharges over the frontal and temporal region – hence he was started on vigabatrin at Day 7 of life.

CONCLUSION

Infants with TSC can be identified early, before the onset of clinical seizures and neurologic sequelae – enabling earlier diagnosis and possibly disease-modifying treatment to improve overall outcome.

Keywords: TSC, Cardiac rhabdomyoma, SEGA, vigabatrin

Abstract ID: A-0015

E-Poster

Fatal cases of urea cycle disorders in male infants - A case series

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BACKGROUND

Urea cycle is a cyclical process involved in hepatic removal of ammonia from the bloodstream. Its disorder is an inborn error of urea synthesis leading to hyperammonemia which in turn leads to irreversible neurological deficit and coma. This disorder has a high mortality rate.

REPORT

Here we report two fatal cases of male neonates with urea cycle disorder.

Baby A was born at term via vacuum-assisted delivery and was complicated with a subaponeurotic hemorrhage which had required treatment with fresh frozen plasma transfusion. He presented in a collapsed state at day 2 of life and had recurrent seizures requiring antiepileptics. Baby B was born at term via spontaneous vaginal delivery and was discharged to mother post-delivery. He was admitted at 13 hours of life for pathological jaundice. He was found to be unresponsive at 65 hours of life. Both infants were ventilated and investigated for possible causes of acute onset of neonatal encephalopathy. Their IEM study confirmed urea cycle defects. Unfortunately, both babies succumbed despite all resuscitative and treatment measures. There were no family history of sudden infant death, recurrent miscarriages or early neonatal death, and they were products of non-consanguineous marriages.

CONCLUSION

Newborns with Urea cycle defects typically appear well at birth and shortly after can present with non-specific signs and symptom. Thus, IEM should always be included in differential diagnosis of any newborn or infants presented with unexplained overwhelming progressive disease.

Keywords: Urea cycle disorders

Abstract ID: A-0016

E-Poster

Newborn atrial flutter in hospital miri - A case report

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INTRODUCTION

Atrial flutter is an uncommon cardiac arrhythmia in newborns and infants. It is usually diagnosed on the electrocardiogram with a fast, irregular atrial activity up to 500 beats per minute and no association with structural heart disease in most of the cases. Synchronized cardioversion remains the most effective way in establishing sinus rhythm.

REPORT

We report a case of a 36 weeks gestation baby born via emergency cesarean section due to fetal tachycardia of up to 210 beats per minute. She was ventilated at birth due to respiratory distress. She started to develop tachycardia at 14 hours of life, ECG has shown supraventricular tachycardia with the presence of P wave and narrow QRS. Echocardiography revealed no structural abnormality. We tried fluid therapy, vagal maneuver as well as anti-arrhythmic medications, including adenosine, amiodarone and digoxin, to which the baby did not respond to. The administration of adenosine resulted in the obvious typical "sawtooth" flutter wave on the ECG. Due to the uncontrolled rate and rhythm, synchronized cardioversion was performed and the rhythm converted to normal immediately. The baby was discharged well after 2 weeks of hospitalization.

CONCLUSION

Neonatal atrial flutter is a rare type of tachyarrhythmia which has a good prognosis. Synchronized electrical cardioversion may be needed in cases that are not responsive to antiarrhythmic drugs.

Keywords: Atrial flutter, antiarrhythmic, synchronized cardioversion, tachycardia, sinus rhythm

Abstract ID: A-0017

E-Poster

Neonatal neuroblastoma with a racing heart: A case report

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BACKGROUND

Neuroblastoma is an embryonal malignancy of the sympathetic nervous system. It is the most common extracranial tumour in infancy. It can occur anywhere along the sympathetic nervous system, including the superior cervical, paraspinal, and celiac ganglia; the majority arise in the adrenal glands. The presentation may vary from asymptomatic or may also be found incidentally from imaging done for other reasons.

CASE

We present a case of thoracic and abdominal neuroblastoma manifesting with supraventricular tachycardia (SVT). A term neonate at 38 weeks gestation was delivered to a mother with an uneventful antenatal history. He was intubated for respiratory distress at birth. Later he developed SVT at 10 hours of life which had required medical treatment. Initial parameters were not suggestive of infection, electrolyte imbalance, or hyperthyroidism. Heart rate remained stable on propranolol. However, a chest x-ray done at day 10 of life for an increase in work of breathing, showed a clear border mass over mediastinum. Bedside echocardiography showed a thoracic mass. A CECT thorax revealed heterogeneous enhancing soft tissue mass at the left lower posterior mediastinum and right retroperitoneal space which was suggestive of a neuroblastoma. The child underwent excision of the tumor, and the HPE had confirmed the diagnosis.

CONCLUSION

Neuroblastoma in infancy commonly presents with compressive symptoms. As it is an embryonal neuroendocrine tumor, originating from neural crest progenitor cells, sympathetic symptoms are possible but rare. High suspicion should prompt further diagnostic workup and intervention as early diagnosis yields a better prognosis.

Keywords: neonatal neuroblastoma, supraventricular tachycardia (SVT), embryonal malignancy

Abstract ID: A-0018

E-Poster

Congenital pulmonary airway malformation (CPAM) with recurrent pneumothorax, a diagnostic conundrum: A case report

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INTRODUCTION

Congenital pulmonary airway malformation (CPAM) is a developmental lung malformation characterized by benign cystic or adenomatoid lung tumors that grow on the terminal bronchioles. The malformation usually occurs sporadically, not hereditary and no association with maternal factors. Infants with this condition may be asymptomatic or can present with hydrops or fetal heart failure, recurrent respiratory infection or pneumothorax..

CASE PRESENTATION

We report a 5-month old baby girl, under our care since birth. She was a term baby intubated at birth for respiratory distress, complicated with persistent pulmonary hypertension of newborn (PPHN) secondary to meconium aspiration syndrome (MAS). She developed recurrent episodes of pneumothorax from day 10 of life, which was initially attributed to MAS. She was ventilated from birth until 2 months old due to recurrent pneumothorax and nosocomial pneumonia. CT Thorax at 1 month old showed right apical pneumothorax and mediastinal emphysema secondary to bronchopleural fistula. Discussion with the paediatric surgical and paediatric radiologist later concluded the possibility of a ruptured CPAM. Child underwent a right upper and middle lobe lobectomy at three months of age. Intraoperative, the right upper and middle lobes appeared emphysematous, with abnormal bullae. Histopathology examination confirmed the diagnosis of CPAM. We weaned her off oxygen on day 16 post-op and the baby was discharged home well.

CONCLUSION

CPAM is often misdiagnosed as a persistent and localized pneumothorax. Therefore, CPAM should be considered in neonates with radio-imaging which are compatible with spontaneous or recurrent pneumothorax. Surgery remains the cornerstone treatment of symptomatic lesions and prognosis is excellent.

Keywords: congenital pulmonary airway malformation, pneumothorax, infant

Abstract ID: A-0019

E-Poster

A case report of neonatal grave's disease

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Neonatal thyrotoxicosis is rare. Most cases are secondary to transplacental passage of thyroid-stimulating immunoglobulin (TSI) from mothers with Grave's disease (GD). The incidence reported in 1-2% of infants born to these mothers. Transplacental passage of maternal thyroid stimulating hormone receptor antibodies (TRAb) binding to fetus Thyroid stimulating hormone (TSH)-receptor on thyroid follicular cells and lead to autonomous thyroid hormone production, can cause in utero and/or postnatal hyperthyroidism. It may last 1-3 months until the maternal TRAb is eliminated from the infant's bloodstream. Even though it is a self-limiting disease, it can be life-threatening and can cause permanent brain damage if the diagnosis and treatment is delayed. We report a female infant who presented with clinical signs of thyrotoxicosis at day 8 of life, with irritability, tachycardia and hyperthermia. She was born to a mother with antenatally undiagnosed Grave's disease but was symptomatic of hyperthyroidism. Her cord TSH level was $< 0.01\text{mIU/L}$. Repeated thyroid function test at day 6 of life showed markedly elevated Free T₄ (FT₄) 66.4pmol/L with suppressed TSH level. She turned euthyroid after a week of Carbimazole. TRAb was detected with a level of 2.35IU/L , confirming the diagnosis of neonatal GD. In conclusion, thyrotoxicosis should be anticipated in high-risk infants with careful clinical and biochemical surveillance in preventing life threatening events.

Keywords: Neonatal Grave's Disease

Abstract ID: A-0020

E-Poster

The outcomes of neonates with COVID-19 positive mothers in University Malaya Medical Centre (UMMC)

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INTRODUCTION

Studies on outcomes of neonates of mothers with COVID-19 are sparse and limited in Malaysia. Given the limited evidence, this study is conducted to explore better evidence-based perinatal care practices.

OBJECTIVES

To determine the outcomes of neonates with pregnant mothers of COVID-19 in University Malaya Medical Centre (UMMC).

SUBJECTS/PATIENTS/MATERIALS

The study was conducted on 72 pregnant mothers with COVID-19 and 71 neonates in UMMC from January until early December 2021.

METHODS

This is a descriptive cohort study with the data obtained from the medical record and analyzed using the SPSS software.

RESULTS

There was a total of 72 deliveries with 71 (98.6%) live births and one (1.4%) still-birth. 48 (66.7%) were term neonates and 24 (33.4%) were preterm neonates. The anthropometric measurement has an average birthweight of 2.793 kg (SD = 0.6333, 95% CI = 2.642-2.944), length of 45.814 cm (SD = 3.897, 95% CI = 44.885 - 46.744) and head circumference of 32.62 cm (SD = 2.634, 95% CI = 31.993 - 33.249). 28 (39.4%) neonates were in respiratory distress and 19 (26.8%) had neonatal sepsis. Out of 56 neonates with SARS-Cov-2 nucleic acid testing using RT-PCR, one (1.8%) was tested positive on day three of life. Overall, 68 (97.2%) neonates were fit during discharge and two (2.8%) passed away due to other underlying medical issues.

CONCLUSIONS

The average anthropometric measurement showed normal values. There is 1 to 3 percent risk of stillbirth, vertical transmission and neonatal death. The study shows 20 to 40 percent incidence of prematurity, respiratory distress and neonatal sepsis. However, other risk factors should be considered including the sample size and the maternal health throughout pregnancy. The majority of the neonates were fit and well during discharge.

Keywords: neonatal outcomes COVID-19, vertical transmission COVID-19, birth outcomes COVID-19

Abstract ID: A-0021

E-Poster

The incidence of SARS-CoV-2 vertical transmission in university malaya medical centre (UMM)

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INTRODUCTION

Coronavirus disease 2019 (COVID-19) is a highly infectious airborne disease caused by severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2) virus, with high risks of severe infection amongst pregnant women. Although recent recommendations suggest low risk of vertical transmission, the lack of consensus opinion for specific practices, such as mode of delivery, skin-to-skin practice, and breastfeeding, warrant further review of the incidence rates for perinatal SARS-CoV₂ acquisition in the newborns.

OBJECTIVES

- To evaluate the incidence rate of COVID-19 vertical transmission in UMMC.
- To investigate the association of perinatal practices with SARS-CoV-2 vertical transmission.

METHODS

A retrospective cohort study of pregnant mothers with positive SARS-CoV₂ RRT-PCR, and of babies delivered between January to October 2021 was conducted at the University of Malaya Medical Centre (UMMC). The newborns underwent SARS-CoV2 RRT-PCR tests on the first day of life and repeated between days-3 to 5.

RESULTS

Of 135 COVID positive pregnant women, 32 women were of category-3 COVID or higher, with one maternal death leading to perimortem Caesarean section. Details of 74 babies, including one set of twins, and one stillbirth were analyzed. 72 (97.3 %) babies tested negative while two were SARS-CoV-2 positive (2.7 %). Both positive cases were delivered to mothers of category 2 COVID-19, via Emergency Caesarean section at 38- and 40-weeks' gestation respectively. They did not receive skin-to-skin, nor were roomed-in following delivery. One baby received mixed expressed breast milk and formula milk, whilst another was on formula feeds from birth till discharge. Infants who were tested negative for SARS-CoV2 were mixed feeding, with two babies roomed in with their mothers from day-3 of life.

CONCLUSION

The risk and incidence proportion of SARS-CoV-2 vertical transmission is very low. Transmission risks are not significantly different between perinatal practice norms for those babies affected.

Keywords: Vertical Transmission of COVID-19

Abstract ID: A-0022**E-Poster****Polyethylene cap as an adjunct in reducing admission hypothermia in preterm infants****Nin Bun San*, Ang Ee Lee***Department of Paediatrics, Hospital Tengku Ampuan Rahimah, Klang, Malaysia***bunsan86@hotmail.com***BACKGROUND**

Hypothermia is defined as body temperature of $<36.5^{\circ}\text{C}$ by the World Health Organisation (WHO). It has been reported previously that up to 55% of preterm infants had hypothermia upon admission to a neonatal unit. Admission hypothermia in preterm infants has been closely related to multiple morbidities and even mortality. Various adjunctive methods have been studied to reduce the incidence of hypothermia.

OBJECTIVE

This study was designed to determine the effectiveness of polyethylene cap as an adjunct to polyethylene body wrap in reducing hypothermia among preterm infants.

METHODS

A prospective study on polyethylene cap as an adjunct in reducing admission hypothermia in all preterm infants below 35 gestational weeks or infants with birth weight less than 1500g admitted to Neonatal Intensive Care Unit (NICU), Tengku Ampuan Rahimah Hospital (HTAR). Syndromic infants or infants with open congenital anomaly lesions were excluded. Infants in both groups were wrapped in polyethylene sheets from the neck downwards immediately after birth without prior drying. A polyethylene cap will be placed immediately in the

study group. On arrival to NICU, axillary temperature was taken immediately in the transport incubator, before being transferred out to NICU incubator.

RESULTS

90 infants included in this study. Median admission temperature to NICU in the infant group with polyethylene cap was higher (36.4°C vs 36.2 °C). Even the difference was statistically not significant ($P=0.20$), however there was lower incidence of admission hypothermia in the infant group with polyethylene cap (25/45; 55.6%) as compared to the infant group without polyethylene cap (31/45; 68.8%).

CONCLUSION

Usage of polyethylene cap as an adjunct to other hypothermia preventive measures reduces the incidence of admission hypothermia among preterm infants. However, in view of small sample size and a single centre study, we recommend a larger prospective multicentre trial in future.

Keywords: Polyethylene Cap in Reducing Admission Hypothermia in Preterm Infants

Abstract ID: A-0023

E-Poster

Congenital pulmonary lymphangiectasia: A case report of management

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BACKGROUND

Congenital pulmonary lymphangiectasia is a rare cause of non-immune hydrops fetalis in neonates. We report a successful management of congenital pulmonary lymphangiectasia.

CASE REPORT

A 33 week and 3 days' gestation neonate with birth weight of 2.83kg was referred at birth for hydrops fetalis. Baby required bilateral chest tube insertion. Further investigations suggestive of non-immune hydrops fetalis.

Baby developed persistent pleural effusion despite chest drain. Pleural fluid analysis concurred with the diagnosis of chylothorax. The pleural fluid turned into chylous drainage on establishing enteral feeding on Day 10 of life. Pleural fluid sent was suggestive of exudative effusion; lymphocytes predominantly, triglycerides (TG)154 mg/dL. TORCHES and Parvovirus screening were negative. Her chromosomal studies were reported as 46, XX. She was then started on octreotide and medium chain triglyceride (MCT) formula (Portagen). By day 53 she was able to tolerate full EBM.

DISCUSSION

Management of congenital chylothorax can be challenging in view of multiple issues such as respiratory failure, increased risks of infection and nutritional deficiency. Prognosis can be guarded particularly with confounding factors of prematurity. Approach mainly directed to effective ventilation, pleural drainage, Somatostatin analog and nutritional support with total parenteral nutrition and medium chain triglyceride formula.

CONCLUSION

Progressive non-invasive approach of drainage, MCT formula and octreotide is initiated prior to decision of surgical intervention, and this approach is often used to reduce lymphatic flow which gradually resolves the lymphatic leakage.

Keywords: neonate, hydrops fetalis, chylothorax, octreotide, Portagen

Abstract ID: A-0024

E-Poster

Neurodevelopmental complications in neonates with severe hyperbilirubinaemia

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BACKGROUND

Hyperbilirubinemia is one of the most common problems during the neonatal period. Severe neonatal hyperbilirubinaemia is a known cause of lifelong neurodevelopmental impairment. Early prompt recognition and timely interventions are imperative for a significant reduction in complications associated with severe hyperbilirubinaemia.

OBJECTIVES

The objectives of this study are to study the neurodevelopmental outcomes of babies with severe hyperbilirubinaemia and to identify its associated risk factors.

METHOD

Neonates admitted to Hospital Sultan Ismail Johor Bahru (HSIJJB) from January 2018 to December 2019 for severe neonatal jaundice and had subsequently completed follow-up up to 2 years old were included in the study.

RESULTS

There were 1,150 babies admitted for severe neonatal jaundice during this two-year study period of which only 90 babies were followed up till 2 years

of age. The study showed that babies who required longer duration of phototherapy were associated with delayed developmental milestones ($p=0.036$) of which speech and language milestones were most frequently affected ($p=0.0437$). Risk factors found to be associated with severe hyperbilirubinaemia were infants of diabetic mothers ($p=0.054$), infants of mothers with blood group O+ve ($p=0.001$), babies with blood group other than O+ve ($p=0.058$) and babies with neonatal sepsis ($p=0.03$).

CONCLUSION

There is risk of neurodevelopmental sequelae in neonates with severe hyperbilirubinaemia, especially in the speech and language milestone. The risk was higher in babies requiring longer duration of phototherapy.

Keywords: Severe Hyperbilirubinaemia, Neurodevelopmental complications

Abstract ID: A-0025

E-Poster

Congenital cytomegalovirus infection in a preterm infant

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REPORT

Congenital cytomegalovirus infection is the most common congenital viral infection and is the leading non-genetic cause of sensorineural hearing loss and an important cause of neurodevelopmental disabilities. We report here a case of a preterm infant at 33 weeks gestation with congenital cytomegalovirus infection presented with multiorgan involvement. Interestingly, the electrocardiogram showed extensive ST depression which could be related to viral-induced myocarditis and his ultrasound brain revealed lenticulostriate vasculopathy over the basal ganglia region. The diagnosis of congenital cytomegalovirus infection was confirmed with urine and blood cytomegalovirus polymerase chain reaction tests. The infant was started on a 6-months course of oral valganciclovir. He tolerated the medication without major side effects and he responded well to the treatment.

Keywords: congenital cytomegalovirus, lenticulostriate vasculopathy, valganciclovir

Abstract ID: A-0026

E-Poster

Risk factors and outcomes of necrotizing enterocolitis in preterm infants: A retrospective case-control study in a tertiary neonatal centre

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BACKGROUND

Necrotizing enterocolitis (NEC) is one of the most unpredictable and devastating diseases in premature infants. The pathogenesis of NEC is multifactorial and has been associated with enteral feedings, bowel ischemia and infectious causes.

OBJECTIVES

Current study aims to describe the incidence, perinatal risk factors and neonatal outcomes of necrotizing enterocolitis for preterm infants in a regional tertiary hospital.

METHODS

This retrospective cohort study examined the medical records of infants born at 24-31 weeks gestation (n=222) in the year 2020 at a regional tertiary neonatal intensive care unit. The incidence, perinatal risk factors and neonatal outcomes of necrotizing enterocolitis were examined and analyzed using SPSS version 27.0.

RESULTS

NEC was diagnosed in 19 (8.5%) of 222 very preterm infants including 14 stage II NEC and 5 stage III NEC. Multivariate binary regression analysis demonstrated that the requirement of inotropic support within the first week of life (OR: 11.073, $P = 0.003$) and anaemia requiring red cell transfusion (OR: 3.304, $P = 0.031$) were the significant predictors of NEC. Administration of empirical antibiotics within the first 72 hours of life was associated with reduced NEC risk (OR: 0.168, $P = 0.011$). With respect to outcomes, the NEC group had a longer duration of total parenteral nutrition ($P < 0.001$), higher rate of intraventricular haemorrhage ($P = 0.056$), higher rate of periventricular leukomalacia ($P = 0.033$) and higher rate of mortality ($P = 0.035$).

CONCLUSION

The incidence of NEC for very preterm infants in the current study is 8.5% which is similar to that of other multicenter studies. Requirement of inotropic support within the first week of life and anaemia requiring red cell transfusion can independently predict the risk of NEC. The increased morbidity and mortality associated with NEC is significant and NEC remains a major health problem among preterm infants.

Keywords: Necrotizing enterocolitis, risk factors, incidence, preterm infants

Abstract ID: A-0027

E-Poster

Are these legs short? - A case report of proximal femoral focal deficiency (PFFD)

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Proximal femoral focal deficiency (PFFD) is a complex birth defect of the upper part of the femur bone which is either malformed or missing, causing one leg to be shorter. This difference causes problems with walking and can stress other bones and joints in the body.

This baby boy was born at 39 weeks via Emergency LSCS for fetal distress with a birth weight of 3.23kg. His mother is 29 years old, Para 2 with no significant antenatal history. He was vigorous however required CPAP support due to episodes of grunting. On examination of the lower limbs, the right leg appeared shorter than the left with a fixed right Congenital Talipes Equinovarus (CTEV). Right and left femur length was 10 cm and 12 cm respectively, the tibial length was 8 cm bilaterally. The muscle bulk over the right lower limb was reduced. The spine was normal and systemic examination was normal. The Ortolani and Barlow test was negative. He was referred to the orthopaedic team and was screened for Developmental Dysplasia of Hip (DDH). Ultrasound of the hip showed mild DDH. He was put on double diapers and the right leg was on serial casting. He was discharged well after 3 days and planned to be reviewed in 1 week.

PFFD is a condition that affects 1 in 200,000 children and can vary in severity. Children with PFFD have other bone and muscle disorders such as malrotation, limb-length discrepancies, fibular hemimelia, joint instability and muscle weakness. The cause of proximal femoral focal deficiency is unknown.

A team of pediatric orthopaedic surgeons, nurses, prosthetists and physical therapists are needed to customise the treatment plan i.e. staged surgeries, limb-lengthening procedures and prosthetics to address the baby's functional defects.

Children with severe PFFD may require a prosthesis to walk, treatment is geared toward improving the baby's functionality with the prosthetics.

Keywords: Proximal femoral focal deficiency, CTEV, DDH

Abstract ID: A-0028

E-Poster

Vertical transmission of COVID-19, how common?

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BACKGROUND

The coronavirus disease 2019 (COVID-19) pneumonia was first reported in Wuhan, Hubei Province, China, in December 2019, it then became a world-wide pandemic as declared by the World Health Organization on March 11, 2020. As of March 2022, there have been more than 3,800,000 confirmed cases and more than 33,000 deaths reported in Malaysia. Vertical transmission of COVID-19 is reported as 1- 3.2%, defined as the transmission of the infectious pathogen from the mother to the fetus during the antepartum intra-uterine periods, intrapartum body fluid contact during childbirth, or through direct contact owing to breastfeeding after birth.

OBJECTIVE

This study aims to look at the number of infants contracted with COVID-19 via vertical transmission.

METHODS

A retrospective study from March 2021 to February 2022 in Hospital Tengku Ampuan Rahimah (HTAR), Klang was conducted on the vertical transmission rate of all infants born to mother tested positive within 10 days from the time of delivery. Nasopharyngeal swab for COVID-19 reverse transcriptase polymerase chain reaction (RT-PCR) was done soon after birth in the neonatal ward, and this was repeated after 48 hours of life if the first swab was negative.

RESULTS

212 infants were born to women tested COVID-19 positive at or near delivery in HTAR, Klang. Out of these 212 infants, 3 infants (1.4%) were tested positive within the first 72 hours of life. Two infants were diagnosed from the first swab, and another from the second swab.

CONCLUSION

Even though vertical transmission of COVID-19 is uncommon, the vulnerability of neonates population and unfamiliarity with this novel virus imposed a great challenge to the health care team.

Keywords: Vertical Transmission of COVID-19

Abstract ID: A-0029

E-Poster

Correlation of risk factors, severity and MRI brain findings in newborns diagnosed with moderate to severe neonatal encephalopathy requiring therapeutic hypothermia in a regional tertiary hospital

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BACKGROUND

Hypoxic ischemic encephalopathy (HIE) is one of the most serious birth complications mainly affecting term infants.

OBJECTIVES

Our study aims to describe the incidence of HIE and risk factors associated with moderate to severe neonatal encephalopathy.

METHODS

Medical records of patients diagnosed with moderate and severe neonatal encephalopathy who underwent therapeutic hypothermia from July 2016 to November 2021 were reviewed in this retrospective cohort study. The incidence, risk factors and MRI brain findings were analyzed using SPSS version 26.0. Binary logistic regression was performed to determine the risk factors associated with severe neonatal encephalopathy.

RESULTS

There were 90 newborns who fulfilled the criteria for therapeutic hypothermia, of which 75 (83.3%) had moderate and 15 (16.7%) had severe HIE based on modified Sarnat staging. The incidence of HIE was 1.5 per 1000 live births. All patients required intubation at birth with median (IQR) Thompson score of 12.0 (5.0) at first hour. The need for cardiopulmonary resuscitation (CPR) increased the odds of severe HIE with adjusted OR (95% CI) of 15.97 (1.86, 137.52). Delivery by caesarean section reduced the odds (95% CI) of developing severe HIE by 80% (29-99%) while every one-point increase in the APGAR score at ten minutes, reduced the odds of severe HIE by half (0.264, 0.819). Out of the 90 newborns, 73 had an MRI brain done, at a median (IQR) age of 11 (3.3) days. 43(58.9%) of them had evidence of HIE while 18(24.7%) had normal MRI's.

CONCLUSION

The incidence of HIE in our centre was comparable to rates reported in developed countries. Need for CPR at birth increased the risk of development of significant HIE, while delivery by caesarean section and good 10-minute APGAR scores, reduced this risk. A significant proportion of the studied newborns (24.7%) were found to have a normal MRI brain.

Keywords: Moderate-severe neonatal encephalopathy, Neonatal Encephalopathy, Hypoxic-Ischemic Encephalopathy, Magnetic Resonance Imaging, Therapeutic Hypothermia

Abstract ID: A-0030

Oral

Prediction of neonatal morbidity and birth defects in diabetic pregnancies in a regional tertiary hospital cross-sectional study

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BACKGROUND

Diabetes mellitus (DM) in antenatal mothers is a major public health issue in Asia. This study aims to describe the incidence of neonatal morbidities and birth defects among infants of diabetic mothers (IDM) born in our center and its associated factors.

METHODS

This is a cross-sectional study conducted at a regional tertiary hospital in Malaysia between November 2021 and February 2022. All neonates born to mothers with diabetes mellitus were enrolled. Exclusion criterion was parental refusal to consent. Logistic regression was performed to determine factors associated with diabetic-related neonatal morbidities.

RESULTS

A total of 170 infants were enrolled in the study of whom 51% were female. One hundred and fifty-one (89.9%) mothers had Gestational diabetes, while 5 (3%) had type 1 diabetes mellitus and 12 (7.1%) had type 2 diabetes mellitus. Morbidities that were observed in IDM include jaundice (50, 29.6%), respiratory

distress (40, 23.5%), hypoglycemia (31, 18.2%) and birth trauma (7, 14.1%). Median (IQR) haemoglobin A1C of mothers with diabetes mellitus ranged between 6.9% (1.1%) to 7.1% (1.2%). Thirty two (76.2%) mothers were on low dose insulin and 25 (14.8%) of them had suboptimal sugar control during pregnancy. Congenital heart disease (CHD) was the most common birth defect among infants of diabetic mothers and logistic regression showed significant associated risk factors were third trimester HbA1c, insulin treatment and type of DM. With every 1% increment in HbA1c, the odds ratio (95% confidence intervals) for CHD doubled (1.13-3.88). Insulin treatment and diagnosis of type 2 DM increased the odds of CHD by 34 (4.24-283.18) and 12 (2.83 – 51.60) times respectively.

CONCLUSION

Infants of diabetic mothers in this cohort had a variety of neonatal morbidities and birth defects. Poor glycaemic control during pregnancy predicts development of these morbidities.

Keywords: birth defect, gestational diabetes mellitus, neonatal morbidity

Abstract ID: A-0031

E-Poster

Congenital infantile fibrosarcoma: A rare cause of a large tongue mass in newborn

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Congenital infantile fibrosarcoma is an uncommon tumour that usually occurs in the first four years of life. Despite its rarity, it is the most common soft tissue sarcoma in infants and constitutes between 20% to 50% of the malignant soft tissue tumours in neonates. To date, the treatment modalities are surgery, chemotherapy and radiotherapy. Distant metastases are rare in Infantile Fibrosarcoma, but local recurrence is common. Infantile fibrosarcoma generally has better prognosis than adult fibrosarcoma with 90% survival at 5 years.

We report an exceptional case of huge fibrosarcoma involving solely the tongue of a term infant with uneventful antenatal history who was delivered via EXIT (Ex Utero Intrapartum Treatment) procedure. A huge oral mass was detected and evaluated during antenatal and postnatal period respectively by magnetic resonance image (MRI) scan. Postnatal MRI revealed a large hypervascular solid tongue mass, causing expansion of the tongue and oral cavity and protruding out of the mouth. The mass involves the entire tongue from the base till anterior tip and normal muscle architecture is not visualised. Doppler ultrasonography noted dilated lingual artery and its branches at the posterior tongue and increased vascularity in the rest of the tongue mass. Open tracheostomy was done in anticipation of airway compromise. Histopathology and immunohistochemistry of tongue mass biopsy done at one week of life showed the spindle cells are positive toward Vimentin, CD99,

Bcl2 and TLE-1 (nuclear, moderate intensity, 50% of cells). The immunohistochemical profile is non-specific. Following histopathology and immunohistochemistry report, 4 cycles of neoadjuvant chemotherapy and surgery were planned. This infant will be closely followed up with serial MRI imaging to determine the outcome.

Keywords: Fibrosarcoma, tongue, infant

Abstract ID: A-0032

E-Poster

Pierson Syndrome: A rare etiology of congenital nephrotic syndrome

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BACKGROUND

Pierson syndrome is a rare autosomal recessive disorder, typically characterized by congenital nephrotic syndrome with diffuse mesangial sclerosis, peculiar ocular anomalies with microcoria and neurodevelopmental deficits. These phenotypes are caused by mutations in the LAMB2 gene, which encodes laminin $\beta 2$.

REPORT

We are reporting a borderline premature boy (35 weeks 1 day) who is the firstborn of non-consanguineous, healthy parents. Antenatally, the mother was detected to have oligohydramnios and the fetus had a unilateral calcified fetal kidney. This index case was born with good Apgar scores, but had poor tone. He later developed worsening oedema, oliguria, and kidney impairment. He was found to have congenital nephrotic syndrome and he had progressed into end stage renal failure within 1 week of age. Bilateral fixed narrowing of pupils (microcoria) was noted. Ultrasound KUB showed bilateral enlarged kidneys with loss of corticomedullary differentiation. He was treated conservatively and had succumbed at day 20 of life due to the end stage renal failure. A referral was made to the genetic team, and the buccal swab done for the whole exome sequencing confirmed the diagnosis of Pierson syndrome. He had a compound heterozygous mutation in the

gene encoding laminin beta2 (LAMB2), and this is likely to be of an autosomal recessive inheritance pattern.

CONCLUSION

We hereby describe a LAMB2 gene nonsense mutation causing severe form of neonatal presentation of Pierson syndrome. There is no specific therapy for Pierson syndrome. The prognosis is poor because of the progressive impairment of renal function and complications of renal failure.

Keywords: Pierson syndrome, microcoria, congenital nephrotic syndrome, LAMB2 gene

Abstract ID: A-0034

E-Poster

Respiratory syncytial virus hospitalization among high-risk premature infants in hospital melaka

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BACKGROUND

Premature infants have been associated with a higher risk of developing respiratory syncytial virus (RSV) infection requiring hospitalization, as well as a higher mortality rate. Palivizumab is used as prophylaxis against RSV infection. This research was conducted with the aim of obtaining data on palivizumab prophylaxis among preterm infants and its associated hospitalization and mortality rate, as compared to those without palivizumab prophylaxis, in a Malaysian context.

OBJECTIVES

To compare the RSV hospitalization and mortality rates among high-risk premature infants with and without palivizumab prophylaxis delivered in Hospital Melaka from 2014 till 2020.

METHODS

This is a retrospective cross-sectional study using a purposive sampling method to recruit the patients. The study was conducted in the paediatric department of Hospital Melaka. Medical records of preterm infants less or

equal to 35 weeks delivered in 2014 till year 2020 were reviewed. Data were analysed using IBM SPSS Statistics version 26.0. Descriptive statistics and Chi-square test were applied, and the statistical significance was set at $p < 0.05$.

RESULTS

Out of the 493 preterm infants recruited, 57.4% were male infants, the Malay ethnic being the majority group (84%) and the predominant age group was 28-32 gestational weeks (48.3%). 8.9% were hospitalized due to RSV infection. 17.6% of the premature infants had underlying bronchopulmonary dysplasia. Those administered with palivizumab prophylaxis had significantly lower rates of RSV hospitalization (5.8%, p value 0.000) as compared to those without palivizumab prophylaxis (9.6%) during the first year after administration. Similarly, the RSV mortality rate was significantly higher amongst those without palivizumab prophylaxis (10.3%, p value 0.000) compared to those with palivizumab prophylaxis.

CONCLUSION

Preterm infants without palivizumab prophylaxis administration demonstrated a significantly higher rate of RSV hospitalization and mortality as compared to those who had been administered with palivizumab prophylaxis.

Keywords: Palivizumab, respiratory syncytial virus, preterm, hospitalization, mortality.

Abstract ID: A-0035

E-Poster

A rare neonatal presentation of incontinentia pigmenti with severe pulmonary hypertension

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BACKGROUND

Incontinentia Pigmenti (IP) is a rare X-linked dominant multisystem disorder characterized by evolving skin lesions following the Blaschko lines. Pulmonary hypertension (PHT) is a severe, lethal, and rarely reported complication. We describe a case of IP in a newborn female, who presented with vesiculopapular skin lesions and persistent pulmonary hypertension of newborn (PPHN) at birth, followed by severe pulmonary hypertension at day 11 of life, and had responded to vigilant pulmonary hypertension management.

REPORT

A term 39 week gestation female infant was born vigorous via caesarean section with birth weight 2.7kg and after an uneventful antenatal history. There was no family history of skin disorders. Few vesicles and papules were noted over the trunk and limbs at birth. At 4 hours of life, she was intubated for respiratory distress. Echocardiography revealed severe PPHN. She responded well to inhaled nitric oxide (iNO), 2 inotropes, intravenous prostaglandin, and was extubated to nasal prong oxygen on day 5 of life. Meanwhile, her skin lesion turned verrucous (Stage 2) with a blaschkoid pattern, suggesting IP. A skin biopsy (day 5 of life) was consistent with the diagnosis (eosinophilic exocytosis and spongiosis). On day 11 of life, she was reintubated for severe pulmonary hypertension, responding poorly to iNO. Sildenafil was added and optimized expeditiously with careful titration of inotropes. She had improved and was successfully extubated on day 21 of life. Meantime,

she developed hyperpigmentation and hyperkeratotic plaques at 3 weeks old. She remained oxygen-dependent and was discharged with home oxygen therapy at 2 months old. Her latest echocardiography (3 months old) revealed a well-controlled PHT, while home oxygen therapy was continued.

CONCLUSION

Pulmonary hypertension is an exceptional and fatal complication of IP with few reported surviving cases among neonates. Our case suggests that vigilant pulmonary hypertension management and efficient use of sildenafil may change this dreadful outcome.

Keywords: incontinentia pigmenti, pulmonary hypertension, persistent pulmonary hypertension of newborn, sildenafil, Blaschko lines, verrucous, hyperpigmentation, hyperkeratotic plaque, inhaled nitric oxide

Abstract ID: A-0036

E-Poster

Acute neonatal hepatitis following asymptomatic COVID-19 infection: An infrequent cause of neonatal hepatitis

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BACKGROUND

Neonatal hepatitis following COVID-19 infection has been reported as a sequela of COVID-19 infection. However, the association between COVID-19 severity and the degree of acute hepatitis is unclear. The natural course and duration of hepatitis following the infection is still unknown. This case illustrates a case of prolonged neonatal hepatitis following asymptomatic COVID-19 infection resembling metabolic liver disease.

REPORT

This female Malay infant was referred to our centre on Day 48 of life for prolonged jaundice. She was exclusively breastfeeding and was thriving well. She was diagnosed with asymptomatic Covid-19 infection on Day 24 of life. The patient remained afebrile and well throughout her quarantine period. Examination revealed hepatomegaly (2cm below the costal margin, soft consistency, smooth surface, regular margin and non-tender). Apart from that, she had no splenomegaly or other evidence of chronic liver disease. She had no dysmorphism, and precordial examinations were normal. Ophthalmological examination found no chorioretinitis or cataract.

Blood tests revealed unconjugated hyperbilirubinemia, raised alanine aminotransferase (ALT), aspartate aminotransferase (AST), gamma-glutamyl transferase (GGT), and lactate. Urine organic acids profile showed mild elevation of 4-hydroxyphenyl acetate and homovanillate (which might indicate

phenylketonuria or certain types of tyrosinemia). Repeated tests over two months showed worsening transaminitis, unconjugated hyperbilirubinemia, persistent hyperlactatemia. Out of expectation, liver enzymes and serum lactate have been down-trending and normalised at five months old.

CONCLUSION

This case illustrates that acute hepatitis might still occur in an asymptomatic neonatal COVID-19 infection patient. Hence, liver enzymes monitoring might be routinely needed regardless of the clinical severity. Furthermore, the complication of acute hepatitis might run a long course before resolution. Larger case series or cohort studies are needed to study the prevalence and natural course of neonatal hepatitis following Covid-19 infection.

Keywords: Acute neonatal hepatitis, neonatal COVID-19 infection

Abstract ID: A-0038

Oral

Iron deficiency without anemia among labouring women in a university hospital, Kuantan, Malaysia

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BACKGROUND

In Malaysia, approximately 40% of pregnant women were reported to be anemic and the majority were due to iron deficiency. However, iron deficiency without anemia (IDWA) is often underdiagnosed. This is an alarming problem because the foetus acquires most of its iron in the third trimester and iron deficiency is linked to neurodevelopmental impairment.

METHODS

A cross sectional study was conducted involving 115 healthy, term pregnant women who delivered in Sultan Ahmad Shah Medical Center @IIUM. Maternal blood for hemoglobin level and serum ferritin was taken prior to labour and cord blood for neonatal serum ferritin was taken at birth. The prelabour hemoglobin was then compared with the booking hemoglobin level which was obtained from the antenatal card. Correlation between maternal and cord blood serum ferritin levels were identified using Pearson correlation analysis.

RESULTS

We found that 49.1% of the labouring women were IDWA while only 7.1% were iron deficiency anemia (IDA). 12.3% of the pregnant women were anemic at booking with the lowest Hb 9.9 (M = 12.09, SD = .97). Prior to labour, 12.2% of them were anemic with lowest Hb 8.4 g/dL (M = 11.76, SD = 1.15). The average level of serum ferritin was 37.75µg/L (range 2.10µg/L - 209µg/L). However, there was no evidence of a significant relationship between serum ferritin of the mother and serum ferritin of the cord, $r(110) = .16$, $p = .11$.

CONCLUSION

Most of the healthy pregnant women had low iron despite normal levels of hemoglobin. While treatment in IDWA is controversial and debatable, recommendations should be made to standardize all pregnant women to be tested for iron status since iron is pivotal for the mother and the first 1000 days of their newborn.

Keywords: Iron deficiency without anemia, labour, nutrition, 1000 days of life

Abstract ID: A-0039

E-Poster

Isolated ascites in neonates: Two cases of different aetiology

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BACKGROUND

Neonatal ascites is a rare and perplexing problem with numerous possible aetiologies.

REPORT

We report two cases of isolated neonatal ascites with differing aetiologies, which we encountered in our district hospital setting. Both cases were detected antenatally as hydrops fetalis hence delivery was facilitated immediately. The babies were intubated at birth due to respiratory distress from gross abdominal distention. Post-delivery, the babies were found to have only isolated ascites with gross hydrocele. Both cases required surgical intervention and were referred to a tertiary centre. In case 1, the baby was noted to be anuric in the first 24 hours of life with acute kidney injury. Imaging studies done were suggestive of posterior urethral valve complicated with grade 3 right vesico-ureteric reflux with perinephric urinoma. Urinary ascites occurred due to high pressure obstructive uropathy. The baby underwent vesicostomy without complications. In case 2, the baby presented with rapidly worsening abdominal distention with persistent greenish Ryles tube aspirates. The presence of air under diaphragm in abdominal x-ray were suggestive of a perforated viscus. The baby underwent emergency exploratory laparotomy on day 2 of life, intraoperatively there was a single perforation found at the terminal ileum. Histological finding confirmed spontaneous neonatal intestinal perforation which most likely occurred in utero. In both cases, the ascites resolved postoperatively and the babies were able to be discharged well.

CONCLUSION

Isolated fetal ascites is a separate entity from hydrops fetalis and has shown to have good prognosis with early identification of the underlying aetiology and subsequent surgical intervention.

Keywords: posterior urethral valve, perinephric urinoma, urinary ascites, spontaneous neonatal intestinal perforation, gross abdominal distension

Abstract ID: A-0040

Oral

Incidence of exchange transfusion and associated factors among neonates with severe hyperbilirubinemia

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BACKGROUND

Severe hyperbilirubinemia can lead to kernicterus and hearing loss, and exchange transfusion (ET) is a last line defence method of reducing bilirubin levels. ET has known complications. Consequently, the incidence of ET has declined in developed countries largely due to improved surveillance of neonates with clinically significant jaundice. In contrast, excessive rates of ET still persist in low and middle-income countries over the years. The aim of our study was to evaluate the incidence of ET in our population, associated factors for those who underwent ET, and the clinical outcomes.

METHODS

A retrospective study was conducted based on a Severe Jaundice Registry from 2017 to 2019. Neonates born ≥ 35 weeks of gestation who were hospitalized for jaundice and underwent ET were included in the study. Severe jaundice was defined as total serum total bilirubin levels exceeding 340mmol/L. We examined clinical characteristics and adverse events after ET.

RESULTS

A total of 8,137 out of 31,501(25.8%) neonates were admitted to our centre for jaundice. 324 (4.0%) of them had severe jaundice. Of these 324, 66

(20.4%) neonates had undergone ET. We found that those who had undergone exchange transfusion presented with early age onset at day 3 of life and higher serum bilirubin on admission ($> 390\text{mmol/L}$). In addition, we found that neonates with ABO incompatibility, neonates with blood group type B had a higher risk of undergoing ET even though their Coomb test was negative. Other associated factors were those with significant weight loss and g6PD deficiency.

CONCLUSION

The results of this study demonstrate that hyperbilirubinemia requiring ET is still high in our population. Important factors are those who have significant weight loss, ABO incompatibility and G6PD deficiency.

Keywords: neonatal Hyperbilirubinemia, exchange transfusion

Abstract ID: A-0042

Oral

Maternal to neonatal transmission of antibody against Covid-19 study - The TRAb CoV-19

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BACKGROUND

The efficacy of vaccination among pregnant women has thus been established. However, fetal protection via vertical antibody transmission during pregnancy remains uncertain. Therefore, we aim to consolidate evidence of transplacental antibody transfer post-maternal vaccination as neonatal protection.

METHOD

A prospective study was conducted with all vaccinated pregnant women with or without a history of COVID-19 infection admitted for delivery at term, were included. Maternal and umbilical cord blood samples were collected within 30 minutes of delivery for quantification of antibodies via ImmusaSAFE® kits and tested for nucleocapsid (N) protein (recent infection) and SARS-CoV-2 (S) (current vaccination) antibodies. Results were considered positive if Anti -N >4634 and Anti -S >3648 based on manufacturer instruction.

RESULT

About 197 mother-baby dyads were included with a mean age of 31.32 years. Most mothers were overweight (26.47 kg/m²) and received mRNA vaccine (Pfizer®) (93.9%). At least 13.2% (n=26) were noted to have recent or past COVID-19 infection. Anti-N antibody was negative for both mother and fetus; Anti-N Ab <4634 (baby: 2642.67, range 1355-5561.42; mother: 2937, range 1681.92-6183, respectively). In contrast, positive antibody (Anti-S antibody) following vaccination was noted to be higher (>3648) in both groups (mother: 17535, range 13533-23000; baby 18349, range 13982-23139). The correlation between maternal and neonatal findings significantly suggested the possibility of transplacental transmission of antibodies from mother to fetus ($p<0.06$). Furthermore, there was a significant rise of anti-N in babies of mothers with past or recent COVID-19 infection ($p<0.001$). Otherwise, either mRNA or live inactivated vaccine had no significant effect on antibody formation ($p>0.05$).

CONCLUSION

Mothers with past or recent COVID-19 or a history of COVID-19 vaccination demonstrated transplacental antibody transmission to the fetus. Hence, maternal COVID-19 immunization should be strongly encouraged as it confers a significant humoral benefit to the newborn, potentially reducing perinatal morbidity and mortality.

Keywords: Covid 19, Vaccination, Transplacental antibody, Immunization

Abstract ID: A-0043

E-Poster

Epidemiology and aetiology of medication administration errors amongst neonates: A systematic review

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INTRODUCTION/BACKGROUND

Medication errors (MEs) may arise throughout the various stages of the medication use process. Although MEs during prescribing are the most common, their likelihood to be intercepted is higher as compared to administration. Hence, medication administration errors (MAEs) are most likely more harmful especially in neonates.

OBJECTIVE(S)

This systematic review aims to critically appraise the evidence on the epidemiology and aetiology of MAEs amongst neonates.

MATERIAL AND METHOD(S)

Nine electronic databases and grey literature were searched for studies without language and publication date restrictions. Studies were included if they were conducted in the Neonatal Intensive Care Unit (NICU) and quantified the MAEs reported or the contributory factors. Study selection, quality assessment, and data extraction were conducted by the lead researcher. Throughout the entire process, all data were double-checked by two independent researchers to ensure reliability, and any differences in interpretation or ambiguous studies

were discussed to reach a consensus. A narrative approach to data synthesis was adopted; data related to error causation were synthesised according to Reason's Accident Causation model.

RESULT(S)

Twenty unique studies were included. The median error rate (IQR) for nine studies reporting total opportunity errors as the denominator for MAEs was 68.0% (41.1% – 86.0%). Amongst the three studies reporting MAEs per neonate, the rate of prevalence was between 17 and 75 MAEs per neonate. Wrong administration technique, wrong drug-preparation and wrong time errors were the three most common types of MAEs. Six studies reported the causes leading to MAEs and found an error-provoking environment to be the most common cause followed by active failures.

CONCLUSION

This is the first comprehensive systematic review gathering available evidence emphasizing the epidemiology and aetiology of MAEs amongst neonates. Important targets such as the common types of MAEs and the reasons for MAEs identified will guide policymakers to implement remedial measures.

Keywords: medication administration errors, prevalence, error causation, systematic review

Abstract ID: A-0044

Oral

Neonatal sepsis in very low birthweight infants: A retrospective cohort study in a tertiary hospital in Malaysia

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BACKGROUND

Sepsis is one of the commonest causes of neonatal mortality. Neonatal sepsis may be classified into early-onset sepsis (EOS) which is defined by positive blood or cerebrospinal fluid (CSF) culture before 3 days of life and late-onset sepsis (LOS) for those after 3 days of life. Data on risk factors, causative organisms and outcome of neonatal sepsis among very low birth weight (VLBW) infants in Malaysia is sparse. This information is important to prevent and improve management of neonatal sepsis.

OBJECTIVE

To describe the incidence, causal organisms, risk factors and neonatal outcomes of neonatal sepsis in VLBW infants in Hospital Tunku Azizah (HTA).

METHODS

Data of all inborn VLBW infants delivered from 1st October 2019 to 30th September 2021 in HTA were collected and reviewed. The data were analysed for incidence of EOS and LOS, causal organisms, risk factors and associated morbidity and mortality.

RESULTS

A total of 406 VLBW infants were included in the study. EOS was found in 0.7% of VLBW infants and LOS in 11.1%. The common organisms that contribute to EOS were *Streptococcus agalactiae* while ESBL- producing *Klebsiella pneumoniae* contribute to 13% of LOS.

Lower birth weight, longer duration of mechanical ventilation, longer duration of central line use and total parenteral nutrition (TPN) use were associated with increased risk of LOS. Prior exposure to broad spectrum antibiotics, which include Cefepime, Meropenem, Vancomycin and Tazocin, were significantly associated with increased risk of LOS.

LOS was associated with increased risk of death, severe intraventricular hemorrhage, necrotising enterocolitis and bronchopulmonary dysplasia.

CONCLUSION

LOS was a significant cause of morbidity and mortality among VLBW infants. Clinical guidelines on prevention and management of neonatal sepsis that are tailored to centre specific characteristics of LOS should be developed.

Keywords: VLBW, neonatal sepsis

Abstract ID: A-0045

E-Poster

The crumpled “Tummy”

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BACKGROUND

Prune belly syndrome (PBS) also known as Eagle-Barrett syndrome or the triad syndrome, is a rare congenital multisystem disease with an estimated incidence of 3.6 to 3.8 per 100,000 live male births. Less than 5% of those affected are female. It is characterised by a triad of absent or deficient abdominal musculature resulting in a prune-like wrinkled abdomen, urinary tract abnormalities, and cryptorchidism. It is also associated with pulmonary, cardiovascular, gastrointestinal and musculoskeletal abnormalities. The exact aetiology of PBS is unknown, but the proposed theories include in-utero bladder obstruction, maldevelopment of the mesoderm, yolk sac defect, and an underlying genetic abnormality.

REPORT

We report a baby girl born at 32 weeks and 2 days of gestation with a birth-weight of 1.42kg. She was born via Caesarean section to a 29-year-old primigravida mother, who had presented in a hypertensive crisis with pre-eclampsia which was complicated by peripartum cardiomyopathy and acute pulmonary oedema. Prenatal ultrasound scan at 30 weeks showed an elongated head with frontal bossing, ventricular septal defect, and bilateral grossly dilated kidneys. Upon delivery, she was intubated for poor respiratory effort and tone. On examination, she had facial dysmorphism, a systolic murmur, a grossly distended wrinkled abdomen, absence of urethral opening, an imperforate anus, and bilateral flat and inverted feet. Echocardiogram revealed a small perimembranous ventricular septal defect. Ultrasound abdomen reported bilateral multicystic dysplastic kidneys with severe hydronephrosis and grossly dilated bowels. In view of multiple congenital anomalies with a very poor prognosis, comfort care was opted, and she succumbed at 41 hours of life.

CONCLUSION

PBS is a rare congenital anomaly of uncertain aetiology and poor prognosis with stillbirths and early neonatal deaths being common. We report a case of female neonate with PBS because of its rarity in females with fewer than 30 cases reported in the literature.

Keywords: prune belly syndrome, neonate

Abstract ID: A-0046

E-Poster

Incidence of failed hearing test and associated factors among neonates with severe jaundice

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BACKGROUND

Up to 60% of term neonates have neonatal hyperbilirubinemia in Malaysia. Reported incidence of hearing loss at the initial testing ranged between 6.7–14.3% at 3 months follow-up. However, apparent medical and economic advantages are yet to be validated in our populations. Therefore, the aim of the study was to retrospectively determine the incidence of abnormal hearing loss and the associated factors among neonates with severe jaundice.

METHODS

This retrospective study is based on the severe jaundice registry, from 2017 to 2019. Severe jaundice was defined as total serum total bilirubin of more than 340 mmol/L. Primary outcome was failed confirmatory hearing loss. The associated risk factor was identified. Hearing screening was done by Automated Auditory Brainstem Response(AABR). Confirmation test was done by auditory brainstem response(ABR).

RESULTS

A total of 8,137 out of 31,501(25.8%) neonates were admitted to our centre for jaundice. Of these admitted, 324 (4.0%) neonates had severe jaundice.

Among all, 24(7.4%) neonates have failed the hearing screening. However, only 1 neonate (0.3%) had confirmed sensory hearing loss. Risk factors that were identified in the guideline did not appear in our study. The proportion of neonates who had undergone exchange transfusion (ET) with failed hearing screening (25%), did not differ significantly from the portion of neonates with normal hearing who had also undergone ET(20%). Those who had undergone ET presented early, from day 3 of life, and with high serum bilirubin levels on admission(>390mmol/L). In addition, 3.6% of those who have prolonged jaundice, which supposedly has a lower risk, can still potentially have abnormal hearing screening, even though all of them eventually passed the confirmatory hearing test.

CONCLUSION

Hearing loss is a known risk for infants with severe neonatal hyperbilirubinemia. However only 1 out of our 324 babies went on to develop sensory hearing loss. Hearing loss does not appear to be a common complication of severe neonatal hyperbilirubinemia in our centre, possibly due to our early intervention strategies.

Keywords: neonatal hyperbilirubinemia, hearing loss

Abstract ID: A-0047

Oral

Risk factors and outcomes of very premature neonates less than 32 weeks with anaemia requiring transfusion

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BACKGROUND

Anaemia of prematurity (AOP) is a common condition with serious consequences among very premature infants. Although its pathophysiology is well-described, the risk factors and minimum blood transfusion threshold are still debatable with no accepted universal consensus.

OBJECTIVES

To compare the risk factors, gestation-specific haemoglobin trends, and outcomes of infants less than 32 weeks gestation who require transfusion, versus those not transfused. We aimed to identify modifiable risk factors to enable implementation of preventive and corrective measures to ultimately reduce blood transfusion needs in this population.

METHODS

A retrospective cohort study was conducted in a tertiary teaching hospital in Malaysia between January 2016 and December 2019. The socio-demographic characteristics, risk factors, and haemoglobin trends were evaluated using univariate and multivariate analysis.

RESULTS

Data of 295 premature infants were analysed, where 135 (45.8%) received blood transfusion. All premature infants experienced variable degrees of anaemia, with mean haemoglobin (Hb) trough levels being 7.4 ± 3 g/dL in the transfusion group. Transfusion group had significantly lower Hb at birth (16.1 ± 3 g/dL, $p < 0.05$), but comparison of the haemoglobin trends between the two groups was not significant. Multiple logistic regression revealed that extremely premature infants, extremely low birth weight, mechanical ventilation days, umbilical catheter insertion, and phlebotomy frequency were significantly associated with transfusion requirement. Premature infants requiring blood transfusion had poorer outcomes where there were delays in achievement of full feeding, prolonged hospital stay, and higher mortality rates compared to those not transfused.

CONCLUSION

Gestational age, birth weight, mechanical ventilation days, umbilical catheter insertion, and phlebotomy frequency affect the requirement for blood transfusion. Ensuring better Hb at birth by possibly delaying cord clamping, and reducing iatrogenic blood loss as a modifiable risk factor will help reduce the requirements for blood transfusion. Infants requiring transfusion have poorer outcomes i.e., bronchopulmonary dysplasia, retinopathy of prematurity and higher mortality rate.

Keywords: Anaemia of prematurity, risk factor, blood transfusion, less than 32 weeks

Abstract ID: A-0048

Oral

Clinical characteristics of COVID-19 in young infants and the protective effect of maternal COVID-19 vaccination and breastfeeding against severe disease

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BACKGROUND

COVID-19 presents in a spectrum with varying severity across different age groups of patients. Young infants may also present with severe disease.

OBJECTIVE

We describe the clinical characteristics of infants hospitalized with COVID-19 and examine the relationship between maternal COVID-19 vaccination and breastfeeding on the outcomes and severity of COVID-19.

METHOD

A retrospective observational study was performed among infants aged 6 months and below who were hospitalized for COVID-19 in Hospital Tuanku Ja'afar Seremban, Negeri Sembilan between 1 February 2022 to 30 April 2022. Demographic, clinical data, breastfeeding practices and maternal vaccination status were extracted from medical records and evaluated against the severity of COVID-19 among infants.

RESULTS

A total of 102 infants were included, 53.9% were males with a median age of 11 weeks old (IQR 5-20). 16 patients (15.7%) had pre-existing comorbidities, including prematurity. Fever was the most common presenting symptom (82.4%), followed by cough (53.9%) and rhinorrhea (31.4%). 41 infants (40.2%) presented with severe disease, defined as pneumonia requiring supplemental oxygen and 2 (2%) required PICU admission. All patients were discharged alive. The median age of infants with severe disease was 13 weeks (IQR 7-22). The proportion of infants who are exclusively breastfed was higher in the non-severe group (37.7%) compared to the severe group (9.8%, $p = 0.002$). The proportion of mothers who were recently vaccinated for COVID-19 was also higher in the non-severe group (47.5% vs 26.8%, $p = 0.036$).

CONCLUSION

Maternal COVID-19 vaccination was associated with lower rates of severe COVID-19 in young infants. Exclusive breastfeeding was associated with lower rates of severe disease. These observations suggest a role for maternal vaccination and breastfeeding to protect young infants from severe COVID-19.

Keywords: COVID-19, infants, risk factors, vaccination, breastfeeding

Abstract ID: A-0049

Oral

The impact of Covid-19 on progression of labour: A single-institution case series report

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BACKGROUND

While COVID-19 has been spreading, management of vaginal delivery in COVID-19 remained unclear due to lack of research focusing on the effects of COVID-19 on characteristics of labour. Lack of information for management of labour have resulted in increasing iatrogenic caesarean section. Our hospital established a multidisciplinary scheme for expectant management.

OBJECTIVES

The objective of our study is to investigate the effects of COVID-19 on vaginal delivery: not only its characteristics but its safety, including vertical transmission.

METHODS

This was a single-institution retrospective study for investigating the effects of COVID-19 on vaginal labour. Pregnant COVID-19 women who transferred and delivered in our hospital from September 2020 to March 2022 were enrolled.

We defined COVID-19-infected women undergoing vaginal delivery during and post-isolation as COVID-19 group (n=50). Control group was defined as non-COVID-19 women undergoing vaginal delivery at term gestation (n=258). With comparison between both groups, we analysed labour time, Bishop score, and COVID-19 antigen from pregnancy-associated materials.

RESULTS

Our main results were as follows: 1) COVID-19 group showed rapid progressing labour on 1st and 2nd stage of labour and after rupture of membrane compared with control group (control vs. COVID-19, min: 545.7 ± 459.1 vs. 291.3 ± 155.2 , $P < 0.01$; 38.8 ± 42.8 vs. 20.9 ± 14.6 , $P < 0.01$; 80.6 ± 104.3 vs. 47.9 ± 50.2 , $P = 0.02$, respectively); 2) COVID-19 group represented higher Bishop score on initiation of labour than control group (control vs. COVID-19; 5.1 ± 2.7 vs. 6.4 ± 2.6 , $P < 0.01$). No vertical transmission was found on conventional COVID-19 testing from pregnancy-related materials.

CONCLUSION

We found the novel characteristics of labour in COVID-19, rapid labour and well-ripening uterine cervix on labour. Systemic inflammation may affect well-ripening cervix and excessive contraction, resulting in rapid progression of labour. Our findings can contribute to further management of labour in COVID-19.

Keywords: COVID-19, labour time, management of labour, perinatal outcome, pregnancy, rapid delivery

Abstract ID: A-0050

Oral

A multi-institutional web-based survey of cardiopulmonary management for extremely preterm infants during the transitional phase in Japan and the UK

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INTRODUCTION

Less invasive surfactant administration (LISA) and neonatologist-performed echocardiography (NPE) are expected to have the potential to reduce morbidities associated with prematurity (i.e., bronchopulmonary dysplasia, haemodynamically significant patent ductus arteriosus [PDA] and intraventricular haemorrhage). However, the prevalence of these new therapeutic strategies may differ worldwide.

OBJECTIVE

This survey aims to evaluate the uptake of LISA and NPE in Japan and the United Kingdom (UK).

MATERIAL AND METHOD

From June to December 2021, we conducted a web-based survey to evaluate the uptake of LISA and NPE in Japan and the UK. The questionnaire consisted of questions about the general information of each NICU, interest in LISA and NPE, and the implementation status of their strategies. In Japan, the

web-based questionnaire was disseminated to the members of the Perinatal Circulation Management and the Japanese Neonatologist Association. In the UK, the members of the British Association of Perinatal Medicine and the Wales Maternity and Neonatal Network were involved in this survey.

RESULTS

Overall, 227 neonatologists (162 institutes) completed the questionnaires: 202 (150) from Japan and 15 (12) from the UK. Only 8% of neonatologists used LISA, and 75% would not plan to start LISA. The main reasons for not implementing LISA were the inexperience of the technique and the doubt about the evidence of LISA. Meanwhile, 89% of neonatologists performed LISA in the UK. Almost all Japanese neonatologists (99%) and a quarter of British neonatologists (27%) performed echocardiography independently. Besides, Japanese neonatologists preferred to intervene in PDA proactively based on the echocardiographic findings, whilst British neonatologists tended to expect spontaneous PDA closure.

CONCLUSION

Most Japanese neonatologists are still indifferent to LISA, as they doubt its advantages. NPE has become commonplace in Japan, but NPE is not a standard modality in the UK.

Keywords: Preterm infant, Less invasive surfactant administration, Echocardiography, Survey

Abstract ID: A-0051

Oral

The combination effect of oxytocin administration and epidural anesthesia in labor for short- and long- term offspring outcome. A scoping review

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INTRODUCTION

Although epidural analgesia (Epi) and oxytocin administration before birth have been investigated, their association with offspring's outcome and they are often used together, the combined effect of them are still unclear.

OBJECTIVE

This scoping review aimed to map current evidence on the outcome of offspring from mothers who were exposed to both Epi and oxytocin in labor.

EVIDENCE REVIEW

We followed the Preferred Reporting Items for Scoping Reviews and Meta-analyses Extension for Scoping Reviews (PRISMA-ScR). All studies were identified by searching MEDLINE/Pubmed, Web of science, Cochrane library and hand searched articles since inception to 30 December 2021. Two independent reviewers screened articles with title, abstract and full-text screening which described the offspring's outcome who were delivered from mothers exposed to both oxytocin administration and epidural anesthesia in labor. Evidence was mapped to several themes that answer the research questions of this review.

FINDINGS

We identified only 7 of 487 studies were eligible for this scoping review. The seven studies were one systematic review (including 2 randomized control trials), 5 cohort studies and 1 retrospective study. Regarding neonatal outcome from birth to one month after delivery, babies from mothers medicated with both Epi and oxytocin showed some differences in behavior or physical assessment when compared to those from mothers with no Epi or no oxytocin. No studies investigated after one month the outcome of offspring from mothers exposed to both Epi and oxytocin in labor.

CONCLUSIONS

This scoping review found all of the included studies focused on offspring's outcome up to 1 month of age who were delivered from mothers exposed to Epi and oxytocin administration before birth and the gap of knowledge after then. Further study is needed in this area which will be helpful and additional information for both women and clinicians considering Epi and oxytocin administration during birth.

Keywords: oxytocin, epidural analgesia, delivery, neurodevelopment

Abstract ID: A-0052

E-Poster

Young-onset non-obese type 2 diabetes in a patient who was born as an extremely preterm small-for-gestational age infant

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INTRODUCTION

Small-for-gestational age (SGA) infants who have restricted growth in utero are at high-risk for type 2 diabetes in adulthood after developing obesity. Furthermore, an early timing of adiposity rebound (AR) at ≤ 4 years of age has been reported to be related to the development of obesity and type 2 diabetes.

REPORT

The patient was a 20-year-old man. He was born by caesarean section at 27 weeks and 4 days of gestation with SGA. His birthweight was 642 g (-2.89 standard deviation). The cause of SGA was due to hypertensive disorders during pregnancy. No early AR or obesity were observed in his infancy, childhood, or school age. He visited his family physician with a chief complaint of pain when urinating. A urine analysis revealed positive urine glucose levels (5+), and he was referred to a diabetologist. At the time of admission (20 years), his height, body weight, and BMI were 169.3 cm, 58.5 kg, and 20.4, respectively. His fasting blood glucose and HbA1c levels were 175 mg/dL and 11.6%, respectively. The insulin response on a glucagon load test and urinary storage connecting peptide immunoreactivity test was normal. The results were negative for anti-glutamic acid decarboxylase antibody and

anti-insulinoma-associated antigen-2 antibody. Based on these results, he was diagnosed with type 2 diabetes. His body composition analyses using InBody s10® (medical device approval number: 223AFBZX00130000) at the onset of diabetes, his body fat percentage and body fat mass were within the normal range (18.4% and 10.8 kg, respectively). However, his muscle mass was 44.7 kg, which was low compared with a standard reference, and he did not exhibit significant fat accumulation.

CONCLUSION

We suggest that patients born as extremely preterm SGA infants should be under careful observation for the development of diabetes, even if they do not become obese.

Keywords: adiposity rebound, body composition analysis, glucagon load test

Abstract ID: A-0053

Oral

Gene expression profile analysis of umbilical cord mesenchymal stem cells revealed fetal programming due to chorioamnionitis

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BACKGROUND

Although chorioamnionitis (CAM) has been demonstrated to be associated with numerous short- and long-term morbidities, the precise mechanisms remain unclear. One of the reasons for this is the lack of appropriate models for analyzing the relationship between the fetal environment and chorioamnionitis and fetal programming in humans. Recent reports suggest that umbilical cord-derived mesenchymal stem cells (UCMSCs) may have the capacity to reflect fetal programming and could be used as an *in vitro* model.

OBJECTIVE

In this study, we aimed to clarify the fetal programming caused by CAM using UCMSCs.

MATERIAL AND METHOD

From nine very low birth weight preterm neonates with CAM (n=4) or without CAM (n=5), we established UCMSCs and subsequently analyzed their gene expression profile and cell function. The diagnosis of CAM was based on clinical and histological findings, and Blanc classification II or severer CAM cases were included in this study.

RESULT

The gene expression profiles obtained by RNA-seq analysis revealed distinctive changes in the CAM group USMSCs. The UCMSCs in the CAM group had a myofibroblast-like phenotype with significantly increased expression levels of myofibroblast-related genes, including α -smooth muscle actin ($p<0.05$). In the pathway analysis, the genes involved in DNA replication and G1 to S cell cycle control were remarkably decreased, suggesting that cellular proliferation was impaired, as confirmed by the cellular proliferation assay and cell cycle assay. Pathway analysis also revealed that genes related to white fat cell differentiation were significantly increased.

CONCLUSION

Our analyses revealed that CAM affected dramatically the characteristics of UCMSCs, which could explain the long-term outcomes of patients who were exposed to CAM and revealed that UCMSCs could be an *in vitro* model of fetal programming affected by CAM. Besides, for future utilization of autologous cell treatment, CAM would affect the outcome of the therapy.

Keywords: UCMSCs, CAM, fetal programing, transcriptome analysis, cell cycle

Abstract ID: A-0054

E-Poster

Oxygenation saturation index as a marker of neonatal hypoxemic respiratory failure

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OBJECTIVE

To investigate the relationship between hypoxemic respiratory failure (HRF) biomarkers, including oxygenation saturation index (OSI) and ratio of percutaneous oxygen saturation (SpO₂) to fraction of inspired oxygen (S/F)

STUDY DESIGN

Between January 2013 and 2020, 77 neonates (gestational age 31.7 ± 6.1 weeks; birth weight 1768 ± 983 g) requiring invasive mechanical ventilation for respiratory disorders were recruited, totaling 1226 arterial blood gas samples. We calculated the oxygenation index (OI), OSI, and S/F ratio from medical records. We performed regression analysis to evaluate the correlation between OI and OSI in all samples, those with SaO₂ $\leq 98\%$, and the first sample obtained after ventilator management. The optimal cut-off point of the OSI and S/F ratio for predicting HRF severity was calculated based on the Montreux definition of neonatal acute respiratory distress syndrome (ARDS); mild ARDS: $4 \leq \text{OI} < 8$, moderate: $8 \leq \text{OI} < 16$, and severe: $\text{OI} \geq 16$.

RESULTS

Overall, OI and OSI showed a positive correlation, which improved when the analysis was limited to samples with $\text{SaO}_2 \leq 98\%$. Receiver operating characteristic curve analysis based on mixed effect modeling using only $\text{SaO}_2 \leq 98\%$ samples showed that OSI and S/F ratio could robustly predict HRF. Furthermore, the optimal cut-off points of OSI and S/F ratio for predicting severe HRF were 7.5 and 201, respectively. Similarly, the optimal cut-off points of OSI and S/F ratio for predicting moderate HRF were 4.5 (98.2%, 95.3%) and 306 (98.2%, 92.2%), and those for predicting mild HRF were 2.5 (95.0%, 91.6%) and 325 (89.3%, 90.8%), respectively.

CONCLUSION

OSI and S/F ratio are useful predictors of neonatal HRF. As evaluation of these parameters allows continuous monitoring of changes in oxygenation in neonates in a non-invasive manner, this approach has the potential for wider clinical application.

Keywords: oxygenation saturation index, ratio of percutaneous oxygen saturation to fraction of inspired oxygen, neonatal hypoxemic respiratory failure, non-invasive markers

Abstract ID: A-0055

Oral

Lower birth weight in newborns with trisomy 18 and esophageal Atresia

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BACKGROUND

The effects of medical and surgical interventions on the survival of patients with trisomy 18 (T18) have been reported, leading to changes in perinatal management and decision-making. In patients with T18 who have esophageal atresia (EA), attending physicians and the family sometimes conflict regarding the use of surgical interventions due to their small body sizes. Here, we hypothesized that T18 newborns with EA have a lower birth weight, which affects the selection of the surgical procedure.

OBJECTIVE

To characterize T18 newborns with EA at our hospital over the last 27 years.

METHODS

T18 newborns delivered at our tertiary NICU between 1994 and 2020 were retrospectively reviewed. Clinical information was extracted from our hospital database, and we then made comparisons between the T18 newborns with and without EA.

RESULTS

A total of 101 newborns with T18 were reviewed, with 33 having EA with tracheoesophageal fistulae (TEF) (Gross type C). T18 newborns with EA presented significantly higher rates of polyhydramnios and threatened preterm labor than those without EA (91 vs. 51%; 79 vs. 52%, respectively). The gestational age and body weight (BW) at birth were significantly lower in T18 newborns with EA than those without EA (35.3 vs. 36.9 weeks; 1,282 vs. 1,626 g, respectively). Among 33 newborns with T18 who had EA, 23 (70%) received surgical treatments. However, only 5 received radical repair for EA and the other 18 underwent palliative surgeries such as esophageal banding or TEF division.

CONCLUSIONS

We found that T18 newborns who had EA were born preterm and smaller, probably due to severe polyhydramnios, which limited the selection of surgical procedures. Although this was a retrospective study at a single center, the observation of a low BW in combination with an accurate prenatal diagnosis of EA are important information when counseling parents of T18 newborns.

Keywords: trisomy 18, esophageal atresia, surgical intervention, body weight

Abstract ID: A-0057

Oral

A Prospective study of respiratory support requirements, bronchopulmonary dysplasia and outcome of premature newborns in a tertiary hospital, Kuala Lumpur.

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INTRODUCTION

Prematurity is the leading cause of death for children under five years old. Current technology and nursing care have increased the survival rate of premature newborns.

OBJECTIVE

We aimed to determine the requirement of respiratory support at birth, outcomes at first discharge from the hospital (or at 40 weeks gestation) and associated factors for bronchopulmonary dysplasia(BPD).

METHODOLOGY

A prospective non-interventional study of newborns delivered between 25⁺⁰ to 33⁺⁶ gestational weeks from 1st July 2020 till 31st July 2021 in a tertiary centre, Kuala Lumpur.

RESULTS

There were 358 premature newborns included. Following the WHO classification of prematurity, there were extremely preterm, very preterm and moderate to late preterm of 43(12%), 138(38.5%) and 177(49.4%), newborns respectively. There was significantly higher need for intubation and ventilation, high-frequency oscillatory ventilation (HFOV), non-invasive ventilation (NIV), oxygen supplementation and surfactant therapy among extreme and very preterm($p<0.001$) newborns. Very preterm newborns developed more necrotizing enterocolitis (NEC), intraventricular haemorrhage (IVH), anaemia requiring blood transfusion, patent ductus arteriosus (PDA) in failure and bloodstream infection compared to moderate to late preterm. The overall mortality rate was 8.9%. Of those who survived to 36 weeks of gestational age, 16% developed BPD. Univariate analysis of factors associated with BPD was low birth weight, intubation, HFOV, NIV, oxygen, surfactant therapy, NEC, IVH, anaemia requiring blood transfusion, PDA in failure, clinical sepsis and bloodstream infection. Multivariate analysis showed that low birth weight, oxygen requirement, anaemia requiring blood transfusion, and PDA in failure were associated with BPD development. The median age at discharge was 36⁺¹ gestational weeks, and the median duration of hospital stay was 30.5 days. About 13.6% of premature newborns required hospitalization beyond 40 weeks of gestational age.

CONCLUSION

In general, premature newborns were associated with high respiratory support at birth with high mortality and morbidities that imposes an economic burden on a country.

Keywords: Newborn, Premature, Bronchopulmonary dysplasia

Abstract ID: A-0058

E-Poster

Calorie-protein intake and growth status in very preterm babies: A preliminary analysis

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INTRODUCTION

Preterm babies need a calorie intake of parenteral 90-120 kcal/kg/day or enteral 110-135 kcal/kg/day (EPSGHAN recommendation) to achieve an in-utero third trimester growth velocity of 15 g/kg/day (AAP guideline). The protein intake should concurrently be adequate at 2.5-3.5g/kg/day from day 2 of life (EPSGHAN) to maintain a protein energy ratio PER of 3.2-4.1 g/100 kcal. The objective of this study is whether very preterm babies receive adequate calories and protein intake, and the impact on their growth.

METHODOLOGY

A prospective cohort study on preterm babies with birth weight ≤ 1500 g and gestation ≤ 32 weeks was conducted in 2020. The calorie-protein intake and body weight were monitored in the first 4 weeks of life. A comparison will be made between infants weighing ≤ 1000 g (G1) and 1005-1500g (G2).

RESULTS

There were 221 babies, 89 in G1 and 132 in G2. The mortality rate was 21.7%. Ten babies had NEC (4.5%), and 5 died. The birthweight was 1074 ± 275 g and gestation 28.9 ± 2.5 g. The age at achieving full enteral feeding was

16.4±9.3days and 115 kcal/kg/day was 23.2±10.3days (longer for G1, $p=0.010$). At 21 days, the calorie intake was 106.0±22.6 kcal/kg/day (less for G1, $p=0.047$). It required 28 days to achieve 3.00±1.1 g/kg/day of protein. The mean PER was less than 2.6 g protein / 100 kcal throughout the 4 weeks. The average z scores were -0.59±0.76 at birth, -2.18±0.73 at 28 days and -2.64±0.73 at 36w postmenstrual age. The mean growth velocity by the exponential method was 5.1±3.6 g/kg/day at 28 days and 8.8±3.1 g/kg/day at 36w PMA.

DISCUSSION

The calorie-protein provision and weight status of very preterm babies are suboptimal even for those 1005 to 1500g birthweight. A greater effort to intensify parenteral and enteral feeding is required to reduce long-term nutritional morbidities in these vulnerable babies.

Keywords: very preterm, calorie-protein intake, growth status

Abstract ID: A-0059

E-Poster

Association between antenatal corticosteroid and severity of Respiratory Distress Syndrome: A single centre experience in East Malaysia

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BACKGROUND

Respiratory Distress Syndrome (RDS) is one of the predominant respiratory diagnoses for newborns in Malaysia but it was not well studied in East Malaysia.

OBJECTIVES

We aim to identify the association between the number of antenatal corticosteroid (ACS) doses received with severity of RDS and effects of rescue dose ACS in newborns in our centre.

METHODOLOGY

Newborns diagnosed with RDS at birth in SWACH between April 2021 and December 2021 were studied. Those with other congenital lung pathology were excluded. ACS was given to mothers based on local protocol. Data on the staging of RDS, number of surfactants required, highest FIO₂ requirements and highest ventilation mode were obtained using a standardized audit form and data was analysed using SPSS v26.

RESULTS

A total 144 newborns with RDS were included with a median gestation of 32.4 weeks (27 - 37.6 weeks) and median birth weight of 1.665kg (0.72 - 3.68kg). The number of ACS did not show any association with the stages of RDS ($p=0.877$), numbers of surfactant required ($p=0.230$), highest ventilation mode ($p=0.739$) and highest FiO_2 ($p=0.072$). It showed association with complication of BPD ($p=0.010$), but further regression analysis was unable to explain the correlation ($p=1.0$, OR 1.0, 95% CI= 0.116-8.637). Newborns that received rescue dose ACS ($n=8$) showed no association with the stages of RDS ($p=0.092$), doses of surfactant required ($p=0.716$), highest ventilation mode ($p=0.791$) and highest FiO_2 ($p=0.592$) compared to those did not receive any doses. They also did not show restricted growth in terms of birth weight ($p=0.76$), length ($p=0.978$) and head circumference ($p=0.25$), compared to those who did not receive.

CONCLUSION

We found no significant association between number of ACS received and severity of RDS in our centre, however this could be due to the small sample size of this study.

Keywords: antenatal corticosteroid, respiratory distress syndrome

Abstract ID: A-0060

Oral

Incidence and risk factors of retinopathy of prematurity in premature neonates in a tertiary neonatal center

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BACKGROUND

Retinopathy of prematurity (ROP) is a vaso-proliferative disorder in premature infants. It remains an important preventable cause of blindness in infants.

OBJECTIVES

This study aims to describe the incidence of ROP amongst premature infants born in Hospital Tunku Azizah (HTA) and its associated factors.

METHODS

This is a retrospective review of all inborn infants screened for ROP in HTA, from 1 October 2019 to 31 October 2021. The screening criteria are premature infants born ≤ 32 weeks gestation and/or birth weight ≤ 1500 g. After subject identification, the medical records of each subject were reviewed from the electronic hospital information system.

RESULTS

During the study period, 104 (23.7%) of the 438 infants who underwent screening had ROP. 10 (9.6%) of the infants with ROP had severe disease and all of them regressed after treatment. ROP was associated with female gender (Odds ratio (OR) 1.59, 95% confidence interval (CI) 1.01-2.44), maternal gestational diabetes mellitus (GDM) (OR 1.54, 95% CI 0.97-2.47), maternal chorioamnionitis (OR 2.54, 95% CI 0.038), late onset sepsis (OR 3.27, 95% CI 1.73-6.16), blood transfusion (OR 4.25, 95% CI 2.67-6.76), significant intra-ventricular haemorrhage (OR 5.53, 95% CI 2.08-14.6), bronchopulmonary dysplasia (OR 5.32, 95% CI 3.31-8.55), increased respiratory support, lower gestational age and lower birth weight. On multivariate analysis, ROP was associated with maternal GDM (OR 1.85), female gender (OR 2.01), lower birth weight, lower gestational age and increased respiratory support.

CONCLUSION

ROP remains an important complication of prematurity. Improvement in both antenatal maternal and neonatal care would help to reduce the incidence of ROP.

Keywords: Incidence, Risk Factors, ROP

Abstract ID: A-0061

E-Poster

A neonate with severe protein C deficiency - clinical manifestations and management

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BACKGROUND

Homozygous protein C deficiency is an extremely rare genetic condition affecting approximately 1 in 400 000 to 1,000 000 live births. It is associated with catastrophic and fatal *purpura fulminans* or thrombotic complications with disseminated intravascular coagulation. Symptoms may develop within a few hours of life or days after birth. Early recognition of such patients may be life-saving.

REPORT

We report a case of a male newborn delivered prematurely at 34 weeks gestation with a birth weight of 2.54kg. He developed spontaneous *purpura fulminans* over both gluteal regions, feet and scalp on day 2 of life. These lesions progressively worsened within the next 24 hours. He also had intracerebral bleed but no midline shifts or obvious neurological deficit. Ophthalmology review showed an intraocular involvement with persistent hyperplastic primary vitreous. Thrombophilia screening noted protein C level of 3.76%. However, the sample was taken after 1 unit of fresh frozen plasma (FFP) and this level is likely to have been lower if taken before the transfusion. He was treated with regular FFP transfusion for protein C replacement as protein C concentrate is not available in Malaysia. Enoxaparin 1.5mg/kg

given twice daily as an anticoagulant. He showed an excellent response to the above measures with complete resolution of *purpura fulminans* within a week. Further imaging did not show any evidence of renal or cerebral vein thrombosis. This patient requires lifelong protein C replacement and anticoagulant.

CONCLUSION

Considering the rarity and potentially irreversible outcomes of severe protein C deficiency, prompt recognition and timely intervention can highly impact prognosis. In countries where the protein c concentrate is not available, FFP combined with enoxaparin can be used with excellent outcome. We report this case to highlight the recognition of early onset neonatal *purpura fulminans* in preventing morbidity and mortality.

Keywords: protein c, purpura fulminans

Abstract ID: A-0062

Oral

Real time ultrasound guided umbilical vein catheterization - The ultracath study

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BACKGROUND

Umbilical vein catheterisation is a safe and effective route for administration of various medications and fluids especially in sick neonates with limited venous access. The appropriate placement of the catheter tip has been conventionally determined using plain X-rays. However, malposition is commonly seen and leads to inadvertent risks, complications and increases radiation exposure.

OBJECTIVES

The aim is to determine the efficacy and feasibility of ultrasound guided umbilical vein catheterization in reducing malposition and radiation exposure in neonates.

METHODS

This is a prospective cohort study done in our neonatal intensive care unit (NICU). Neonates admitted from 1/6/2021 – 27/4/2022, who require umbilical vein catheterization and fulfilling the inclusion criteria were selected. They were divided into 2 groups, the intervention (ultrasound-guided) and control (conventional) group. Ultrasound was performed by medical officers who were trained for this intervention purpose. Data was then tabulated and analysed.

RESULTS

119 umbilical vein catheterization were included fulfilling the inclusion criteria with a mean gestational age of 34 weeks and birth weight of 2.14kg. The proportion of malposition in the intervention group was 14.5% compared to 57.8% in the control (RR 0.25, 95% CI 0.1238 - 0.5152, $p=0.0002$). The proportion of increased radiation exposure in the treatment group was 14.5% compared to 55% in the control group (RR 0.27, 95% CI 0.1297 - 0.5434, $p=0.0003$). These results are very encouraging and implies the beneficial role of ultrasonography.

CONCLUSION

Ultrasound is a reliable modality in detecting malposition and preventing its complications. It reduces multiple cannulations and radiation exposure to neonates. This intervention is cost-effective, easily replicable, and the physicians can be easily trained to do it. The findings of this study facilitates the usage of interventional ultrasonography in our daily practice.

Keywords: umbilical vein catheterization, ultrasound guided, neonates

Abstract ID: A-0063

E-Poster

Congenital cytomegalovirus infection: A case report on a symptomatic infant with mega cisterna magna

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BACKGROUND

Congenital cytomegalovirus infection (cCMVI) is one of the most common congenital infection that cause significant morbidity and mortality. The data on prevalence of symptomatic cCMVI in Malaysia is still limited. We reported an infant with incidental finding of mega cisterna magna on prenatal scan who turned out to have symptomatic cCMVI.

CASE REPORT

A baby boy, born at 38 weeks with a birth weight of 2.16kg, was referred at birth for mega cisterna magna. Detailed scan at 25 weeks showed mega cisterna magna. Amniocentesis at 28 weeks showed normal karyotyping of 46, XY. Mother's TORCHES screening reported as CMV IgG positive while IgM negative. MRI Fetus at 34 weeks showed mega cisterna magna and colpocephaly.

Tests for TORCHES sent on day 2 of life were positive for CMV IgM. CMV (PCR) was also detected in the urine. Postnatal MRI Brain on Day 26 of life showed presence of mega cisterna magna and periventricular calcification. Hearing assessment (AABR) showed hearing loss bilaterally. Baby was started on IV Ganciclovir on Day 28 of life for 6 weeks. He was discharged with oral Valganciclovir for a total duration of treatment of 6 months.

DISCUSSION

Majority of children with cCMVI are asymptomatic at birth. Finding of mega cisterna magna during prenatal scan should warrant screening for cCMVI. After birth, detailed clinical examination and timely investigation are important in establishing the diagnosis. Treatment should be discussed with the parents, and to be commenced within the first 30 days of life to improve hearing and neurodevelopmental outcome.

CONCLUSION

Presence of mega cisterna magna in prenatal scan should prompt an investigation for congenital infection such as CMV. Treatment may be offered to symptomatic patients after balancing the risks and benefits.

Keywords: neonatal, congenital infection, cytomegalovirus, prenatal, mega cisterna magna

Abstract ID: A-0064

E-Poster

Prescriptions of anti-rheumatic drugs for women of childbearing age in Japan: An analysis of real-world data from nationwide claims-based database

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BACKGROUND

Rheumatoid arthritis (RA) develops frequently at reproductive age and may require treatment with one or several disease-modifying antirheumatic drugs (DMARDs). However, some DMARDs have been associated with increased teratogenic risk, and newer agents lack safety information during pregnancy.

OBJECTIVE

This study aims to evaluate the prevalence of RA among women of reproductive age, prescriptions of DMARDs including biologics in recent years, and fertility of women with RA, using real-world data (RWD) in Japan.

MATERIAL AND METHODS

This is a retrospective cohort analysis using a nationwide claims-based database in Japan (JMDC). We used the data of 2,645,758 women 15–49 years of age. Women who had at least one prescription for a specific anti-rheumatic drug with Anatomical Therapeutic Chemical Classification System (ATC) were considered RA-affected women.

RESULTS

There were 7,536 (0.28%) women prescribed DMARDs and 2,638,212 women were not prescribed. A year-by-year analysis shows that methotrexate was the most common (35%), followed by prednisolone (22%), salazosulfapyridine, tocilizumab, and etanercept among women with RA in 2018. Biologics have been increasing in recent years as a prescribing trend. When comparing RA and non-RA women in terms of getting pregnant, RA women were less likely to be pregnant (412 (5.5%) vs. 166,337 (6.3%), $p=0.0028$, Odds ratio 0.859, 95%CI 0.77–0.94)).

CONCLUSION

RA women were significantly less likely to be pregnant. In this study, we examined the prescription of DMARDs in women of childbearing age and compared the presence of pregnancy in RA and non-RA women. Further research is needed to better understand how to treat RA before, during, and after pregnancy.

Keywords: Pregnancy, Rheumatoid arthritis, Anti-rheumatoid drugs, Prescriptions, Real-world Data

Abstract ID: A-0066

E-Poster

Probiotics for prevention of necrotising enterocolitis in preterm infants

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BACKGROUND

Necrotising enterocolitis (NEC) affects predominantly preterm infants.

OBJECTIVE

We aim to evaluate the use of probiotics in the prevention of NECs in pre-term infants.

METHODS

This is a retrospective case-control study done in NICU of Sabah Women and Children Hospital, Kota Kinabalu from July 2018- June 2020. The study population consists of all preterm infants with very low birth weight (less than 1.5kg) or gestation less than 32 completed weeks.

Our centre introduced the use of multi-strain probiotics (HEXBIO) from June 2019, based on the criteria mentioned above.

Patients' data were extracted using the Malaysian National Neonatal Registry (MNNR).

The study population was divided into 2 groups: (1) the control group - before the use of probiotics and (2) the group with probiotics. The risk factors

associated with NECs among these 2 groups were evaluated, including maternal age, intrapartum antibiotics, birth weight, as well as co-existing prematurity complications such as RDS and PDA. The main outcome, which is the incidence of NEC stage ≥ 2 (according to Bell's criteria), is then compared. Statistical analysis was done using SPSS Version 26.

RESULTS

242 preterm infants were identified in the control group, and the incidence of NECs is 11 (4.5%). A total of 255 preterm infants were recruited in the probiotics group, and the NECs cases comprise of 6 (2.4%).

These 2 groups shared a similar background in terms of the possible factors affecting the risk of NECs.

As a result, there is an absolute risk reduction of 2.1% and this yields a relative risk of 0.5 (2.4%/4.5%), which means the chance of NEC is twice as likely to occur without probiotics.

CONCLUSIONS

This study shows an overall preventive effect of probiotics on NEC in preterm infants. Further large-scale placebo-controlled trials are still needed to provide conclusive evidence.

Keywords: NEC, prematurity, probiotics

Abstract ID: A-0067

E-Poster

Acute myocarditis complicating RSV infection in infants

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INTRODUCTION

Acute myocarditis is not common in infants but it is associated with severe morbidity. Infection by a virus, especially Coxsackievirus B has been the most widely reported. Here we report 3 cases of myocarditis associated with RSV infection within a six-month period.

CASE PRESENTATION

Three premature babies that were awaiting weight gain were diagnosed to have RSV infection when they developed 'pertussis-like cough'. RSV was confirmed to be present in nasopharyngeal aspirates. Myocarditis was suspected due to persistent tachycardia and the diagnosis was confirmed by abnormal cardiac function on echocardiography and elevated serum Troponin T levels. All 3 babies developed ARDS and required multiple inotropic supports. Intravenous immunoglobulin was given. All 3 babies survived the event.

CONCLUSION

Acute myocarditis needs to be suspected in infants with a viral illness who developed unexplained tachycardia.

Keywords: Respiratory Syncytial Virus, myocarditis, neonate

Abstract ID: A-0068

E-Poster

Rare cause of absent pulses in the lower limbs

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INTRODUCTION

Thrombotic complications have been well described with the use of umbilical arterial lines, but fatal complications are rare.

REPORT

A baby with Down syndrome was treated in a peripheral hospital for meconium aspiration syndrome with persistent pulmonary hypertension of the neonate. On day 5 after birth the baby was referred to our tertiary centre for the management of acute kidney injury with anuria.

Upon admission the pulses in the lower limbs were not palpable and an urgent echo showed that there was no coarctation or interruption of the aortic arch. Ultrasound Doppler revealed a very extensive thrombosis of the descending aorta. Retrospective review revealed that the umbilical artery catheter that was initially inserted had coiled in the lower part of the descending aorta. It had been removed and a new umbilical artery catheter had been inserted on day 1 after birth.

After removal of the catheter, the baby received treatment with a thrombolytic agent and peritoneal dialysis. The baby was not stable enough for thrombectomy.

The baby developed multi organ failure and when the condition deteriorated parents were allowed to hold the baby during the final moments of the baby's life.

CONCLUSION

Fatal complications of umbilical arterial lines are rare. This baby developed extensive thrombosis of the descending aorta and succumbed to multi organ failure.

Keywords: Umbilical arterial catheter, thrombosis, descending aorta, neonate

Abstract ID: A-0069

E-Poster

Does protocol miconazole administration improve mortality and morbidity on surgical necrotizing enterocolitis?

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BACKGROUND

Our previous study reported that miconazole (MCZ) had anti-inflammatory effects and prevented the development of gastrointestinal perforation in premature infants. The purpose of this study was to investigate the potential favorable effect of MCZ administration on necrotizing enterocolitis (NEC) in premature infants.

METHODS

Out of 1172 premature infants, 15 patients with NEC (1.3 %) underwent surgery between 2011 and 2020. Protocol MCZ administration for 3 weeks was applied for neonates born at <26 weeks' gestation or weighing <1000 g. We compared MCZ (+) with MCZ (-) regarding background characteristics, clinical outcome and neurological prognosis using the Kyoto Scale of Psychological Development (Developmental Quotient: DQ). DQ included three domains as follows; postural-motor domain, cognitive-adaptive domain, language-Social domain.

RESULTS

MCZ (+) NEC patients [gestational age: 25 (23-26) weeks, birth weight: 665 (565-781) g] had significantly earlier birth and lower birth weight comparing with MCZ (-) NEC patients [gestational age: 29 (27-30) weeks, birth weight: 1004 (944-1429) g]. MCZ (+) NEC patients (29 [25-30] day) underwent surgery 10 days later comparing with MCZ (-) NEC patients (19 [16-21] day). Body weight at surgery of MCZ (+) NEC patients were almost same as those of MCZ (-) NEC patients in spite of smaller birth weight ($p = 0.142$). Mortality had no significant difference between NEC patients with and without MCZ ($p = 0.600$). There were no significant differences of DQ of all domains at 1.5 years of corrected age [MCZ (+): 79 [58-93], MCZ (-): 91 [87-96], $p = 0.248$] and at 3 years of chronological age [MCZ (+): 84 [56-87], MCZ (-): 86 [85-87], $p = 0.374$], respectively.

CONCLUSION

Protocol MCZ administration did not improved mortality, but it delayed the onset of NEC in neonates born at <26 weeks' gestation or weighing <1000 g and thus kept the DQ level of those patients.

Keywords: miconazole, necrotizing enterocolitis

Abstract ID: A-0070

E-Poster

A descriptive study of neonatal encephalopathy (NE) in the neonatal intensive care unit of seri manjung hospital in Perak, a district hospital in Malaysia

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INTRODUCTION

Neonatal encephalopathies, especially those associated with Hypoxic Ischaemic Encephalopathy(HIE), still occur despite advances in neonatal care. There's little literature on the incidence and outcome after hospital discharge in the local setting.

OBJECTIVE

1) To estimate the incidence of moderate to severe HIE. 2) To estimate the percentage of moderate to severe HIE infants with MRI changes and developmental delays.

METHODS

All infants born at 35 weeks' gestation and above in 2021 diagnosed with neonatal encephalopathy were identified. HIE was classified as mild, moderate, and severe. Incidence of those with moderate to severe HIE was estimated.

RESULTS

In 2021, 4000 live births were born in HSM. 52% of them were admitted to the neonatal intensive care unit for various neonatal conditions that required admission. A total of 17 of these admissions met the criteria for NE. The overall incidence of moderate-to-severe HIE was 3.5 per 1000 live births and one death. Two infants had evidence of a perinatal event. MRI changes and developmental delays were documented at follow-up in 23% of infants with moderate-to-severe HIE.

CONCLUSION

Moderate-to-severe HIE is uncommon. However, these infants require intensive care during therapeutic cooling. Those with MRI changes and developmental delays require long-term medical care.

Keywords: Live births, therapeutic cooling, moderate to severe HIE, developmental delay

Abstract ID: A-0071

Oral

Exploring the use of telemedicine online consultation for premature infants less than 34 weeks: A tertiary centre experience

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BACKGROUND

We aim to understand the experiences of caretakers and doctors when medical consultation among premature infants less than 34 weeks was conducted online using a specifically designed clerking template as a guide for the online consultation of premature infants.

OBJECTIVE

To understand the experiences of caretakers and doctors with online consultation in premature infants less than 34 weeks. A qualitative study.

METHODS

A clerking template was designed for guidance. Recruitment was done in the paediatric clinic of Hospital Universiti Sains Malaysia, Kelantan, from June 2021 until July 2021. Eligible caretakers were contacted via WhatsApp video call by doctors. Questionnaires with open-ended questions were given to the caretakers and doctors after the online consultations.

RESULTS

Of 19 caretakers who were identified, 12 participated, and the online consultations were conducted by four doctors. Developmental assessment manoeuvres were performed, and it took between 19 to 34 minutes (mean of 26 minutes) for a video call session to be completed. The usage of a phone holder or tripod facilitated the session, especially when one parent was in the session. The majority of the participants were able to perform developmental assessment manoeuvres during the online consultation sessions with guidance and demonstration by attending doctors using a manikin. A post-consultation evaluation session showed satisfaction of both participants, i.e. caretakers and doctors.

CONCLUSION

Telemedicine online consultation may be an alternative medium for neonatal developmental follow-up, and it may be feasible in Malaysia by using the WhatsApp application, especially during times of crisis, like the COVID-19 pandemic.

Keywords: consultation, development assessment, experience, premature infant, telemedicine

Abstract ID: A-0072

E-Poster

Retrospective analysis on implementation of kangaroo mother care in hospital miri NICU

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INTRODUCTION

Preterm birth rate in our centre makes up 10% (n=487) in 2019 and 11% (n=491) in 2020 of the live births, with mortality rate of 3% for preterm infants less than 33 weeks. According to WHO and Cochrane review (2016), Kangaroo mother care helps to reduce mortality, nosocomial infection, hypothermia, and improved growth and exclusive breastfeeding.

METHODOLOGY

Kangaroo Mother Care (KMC) Project was introduced in 2020 in Hospital Miri NICU as part of a quality improvement project. Stable preterm infants with postmenstrual age 30 weeks to 34 weeks 6 days were enrolled with mother's consent into the project.

RESULT

A total of 41 infants with the gestation of 32 to 34 weeks 6 days participated, 22 (53.7%) with majority of 41.5% aged 34 to 34 weeks 6 days post menstrual age at the time of enrolment. Mean length of stay was 38.34 days (SD:24.4), time taken to achieve birth weight was 11.4 days (SD: 4.05). Time taken to initiate breastfeeding ranges from 8 to 14 days to >22 days of life, mean: 24.78. Eighteen infants (43.9%) achieved exclusive breastfeeding on discharge. Mother's mental health, knowledge and experience were measured using Likert scale with the total score of 15 for mental health and 18 for knowledge and experience. For mental health score, pre-KMC median score:14, post-KMC median score was 15. There was improvement in the mother's experience upon discharge (p-value: <0.001). For overall experience, the median was 18 with the mean score of 16.88 (SD:1.56).

CONCLUSION

Our study was suspended prematurely as per local pandemic control guideline. Knowing about the benefit of KMC to both mother and infants, we suggest that it should be encouraged and continued with adaptation and modification of the procedure during COVID-19 pandemic.

Keywords: Kangaroo Mother Care, Hospital Miri

Abstract ID: A-0073

E-Poster

Lissencephaly due to fumaric aciduria: A rare entity

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Fumaric aciduria is an autosomal recessive metabolic disorder characterized by early onset but non-specific clinical signs: hypotonia, psychomotor impairment, convulsions, respiratory distress, feeding difficulties and frequent cerebral malformations, along with a distinctive facies. This rare entity is caused by mutations in the Fumarate hydratase (FH) gene.

Our case highlights the course of an Asian boy who was born via vaginal delivery at 36 weeks gestation with a birth weight of 2.46kg. Antenatal scans since the second trimester detected bilateral ventriculomegaly with dilated third ventricle and smooth sulci and gyri. Amniocentesis for fetal karyotyping, microarray and TORCHES screen were unremarkable. Postnatally, he was dysmorphic with a broad forehead, hypertelorism and ectopic anus. Initial MRI brain showed Type II lissencephaly with ventriculomegaly and corpus callosum dysgenesis. Other organ abnormalities included perimembranous ventricular septal defect with patent ductus arteriosus. He was initially supported with non-invasive ventilation, but subsequent progress was stormy with recurrent pneumonias, and required invasive ventilation by Day 15 of life. At 1 month of life, he developed conjugated hyperbilirubinaemia, and metabolic testing with urine organic acid showed a large peak of fumaric acid. Whole exome sequencing then detected a pathogenic FH mutation, and a mutation variant-of-unknown significance (VUS). This led to the probable diagnosis of fumaric aciduria.

Further attempts to wean off ventilator were unsuccessful due to profound apnoea and irregular breathing efforts, and multiple acute life threatening events requiring cardiopulmonary resuscitation that was attributed to

autonomic dysfunction. In view of the grave prognosis, withdrawal-of-ventilation was pursued after discussion with the parents, and the patient succumbed on Day 89 of life. Thus, our case shows the value of considering this rare diagnosis in a dysmorphic patient with abnormal brain gyration and sulcation.

Keywords: neonatology, fumaric aciduria, lissencephaly

Abstract ID: A-0074

Oral

The Use of Amplitude-Integrated Electroencephalography (aEEG) in a neonatal intensive care unit, a single center study

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INTRODUCTION

Amplitude-Integrated Electroencephalography (aEEG) has been used increasingly in neonatal units to enhance neurological monitoring of patients. In the Neonatal Intensive Care Unit (NICU) it is particularly useful in monitoring patients with neonatal encephalopathy and neonatal seizures. Using a 4-channel recording of raw EEG data in a compressed time format, it captures background electrocortical activity, presence of sleep-wake cycling and electrographic seizure activity, which may not be clinically evident. Neonatologists interpret these recordings at bedside to aid in treatment and prognostication of affected patients. Complex cases are referred to neurologists for conventional Electroencephalography (cEEG).

OBJECTIVES

To describe the use of aEEG monitoring in an NICU setting as well as to review the indications, technical adequacy and interpretations of the aEEG recordings in a neonatal unit.

METHODOLOGY

Retrospective analyses of aEEG data recordings as well as corresponding clinical data done in the NICU of Hospital Tunku Azizah from January 2020 to December 2020 were reviewed.

RESULTS

Thirty-eight ($n=38$) patients had aEEG monitoring in the year 2020. Indications were neonatal encephalopathy (17, 44.7%) and neonatal seizures (21, 55.3%). 11 (64.7%) neonates with neonatal encephalopathy received cooling therapy. The causes for the 21 neonatal seizures were meningitis (7, 33.3%), electrolyte imbalances (4, 19.1%), severe intraventricular hemorrhage (3, 14.3%) and seizure disorders (7, 33.3%). 16 (42%) patients had both clinical and electrographic seizures. All 16 patients did not have clinical seizures following anti-epileptic therapy. However 8 (50%) patients had persistent electrographic seizures.

CONCLUSION

The use of aEEG in a neonatal unit is imperative in monitoring and management of patients with high risk of cerebral dysfunction. Improving the skills and knowledge of clinical practitioners in NICU's on the use and interpretation of aEEG will further improve the standard of care of patients.

Keywords: Amplitude-Integrated Electroencephalography, aEEG, neonatal encephalopathy, neonatal seizures

Abstract ID: A-0075

E-Poster

Transient functional hypoparathyroidism in infant of diabetic mother

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BACKGROUND

Hypocalcaemia, hypomagnesemia and hyperphosphatemia are known metabolic complications in infants of diabetic mothers (IDM). These problems are related with functional hypoparathyroidism in these infants.

REPORT

We report a case of an infant of a diabetic mother who was initially admitted for being a macrosomia baby for observation. During the first 24 hours, she was well with no evidence of metabolic complications, both clinically and by lab results. She was discharged to her mom. She was readmitted at day 3 of life for neonatal jaundice and was noted to have hypocalcaemia, hypomagnesemia and hyperphosphatemia. The hypocalcaemia and hyperphosphatemia responded poorly to the usual treatment of calcium and magnesium corrections, and the use of calcium carbonate as the phosphate binder. However, after starting her on a megadose of Vitamin D, her hypocalcaemia and high phosphate slowly normalised.

CONCLUSION

This case highlights the metabolic complications of infants of diabetic mothers. The IDM needs regular electrolyte monitoring especially if the mother's

diabetic control was poor. Educating the diabetic mom during pregnancy is very important to reduce the risk of related complications in the newborns.

Keywords: Hypocalcaemia, Hyperphosphatemia, Hypoparathyroidism, Infant diabetic mother(IDM)

Abstract ID: A-0076

E-Poster

Congenital chylothorax: A rare neonatal cause of non-immune hydrops fetalis

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BACKGROUND

Congenital chylothorax (CCT) is a rare cause of non-immune hydrops fetalis (NHF). We described a case of prenatally-diagnosed NHF which manifested as CCT at day 14 of life after introduction of enteral formula. Our case responded well to pleural drainage, parenteral nutrition, medium chain triglycerides (MCT) based enteral formula, and intravenous octreotide.

REPORT

A preterm 34 weeks gestation male infant was diagnosed with hydrops fetalis (HF) on prenatal scan. There was no family history of hematological or genetic disorder. On the next day, he was delivered via cesarean section for fetal distress with birth weight 3kg. During neonatal resuscitation, the infant was unable to be ventilated hence bilateral pleural tapping was promptly performed to decompress the tension pleural effusion, draining 50cc haemorrhagic fluid. In NICU, he had required high frequency oscillatory ventilation (HFOV) with 2 inotropic supports. Bilateral chest tubes were inserted to drain the massive pleural effusion. His initial pleural fluid analysis was transudative in nature whilst investigations for NHF were negative. He first started enteral feeding on day 10 of life. However, 4 days later, his chest tubes drained chylous fluid. A raised pleural fluid triglyceride (2.1mmol/L) pointed towards chylothorax. Total parenteral nutrition (TPN), MCT- based enteral formula, and intravenous octreotide infusion were started on day 10, day 16, and day 26 of life respectively. He responded well to the combination of therapy.

On day 32 of life, chylothorax resolved and he was extubated 2 days later. After 2 weeks of intravenous octreotide and 6 weeks of MCT- based enteral formula, he was rechallenged with term milk formula. At 3.5 months old, he was discharged home well without re-accumulation of chylothorax.

CONCLUSION

Congenital Chylothorax is rare and lethal. Prompt decompression of pleural effusion, early recognition of chylothorax, and combination therapy are the key to our baby's survival.

Keywords: Congenital Chylothorax, Hydrops Fetalis, Pleural Effusion, Pleural drainage, Total Parenteral Nutrition, Octreotide

Abstract ID: A-0077

E-Poster

Electrolyte derangement in hypoxic-ischaemic encephalopathy: Incidence, predisposing factors, and discharge outcomes

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INTRODUCTION

Management of infants with hypoxic-ischaemic encephalopathy(HIE) requires close monitoring of electrolyte derangements. Sentinel events such as abruptio placenta cause primary energy failure, consequently leading to failure of Na/K pumps causing cerebral oedema, and microvascular brain damage. Early detection and prompt treatment of such derangement may improve HIE outcomes.

AIMS

a)To determine the incidence of serum sodium, potassium, and calcium derangements in patients with HIE within the first three days of life, b)To determine the predisposing factors associated with electrolyte disturbances in HIE patients, c)To compare electrolyte trends between 3 HIE categories, and, d)To determine the association between initial electrolyte disturbances and HIE discharge outcomes.

METHOD

This was a retrospective study conducted over 11 years between January 2011 and December 2021. HIE patients with gestation age ≥ 35 weeks were recruited, where electrolyte status from days 1, 2, and 3 were analysed with

descriptive analyses conducted to summarize the HIE incidence, maternal, and infant characteristics.

RESULTS

The HIE incidence rate between the years 2011 to 2021 was 0.81 per 1000 live births. A total of 48 of 65 patients (73.8%) had sodium abnormalities, whilst 44 (67.7%) and 19 (29.2%) had potassium and calcium abnormalities respectively. The lowest sodium levels were seen in patients with severe HIE. Although overall electrolyte disturbances were not associated with adverse discharge outcomes in patients with HIE, there was a significantly higher proportion of patients with abnormal sodium who died (83%). All severe HIE patients who required antiepileptic medications upon discharge had abnormal sodium.

CONCLUSION

Electrolyte disturbances such as hyponatremia, hypokalemia, and hypocalcemia are common in HIE patients, with mean levels of sodium lowest in patients with severe HIE. Low sodium levels may be associated with adverse discharge outcomes in patients with severe HIE. Therefore, vigilant monitoring of electrolytes during the management of infants with HIE should be routinely recommended.

Keywords: hypoxic ischaemic encephalopathy, electrolyte derangement, hyponatraemia, sentinel event

Abstract ID: A-0078

Oral

Predictors of neonatal organ dysfunction sepsis scores for mortality and illness severity in late onset neonatal sepsis

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INTRODUCTION

Neonatal sepsis remains a leading cause of infant mortality. Prediction and detection of sepsis is difficult, relying on non-specific clinical signs and sepsis markers. Various scores have been used to monitor disease severity and predict neonatal mortality.

METHOD

This retrospective cohort study conducted over a 5-years period in a tertiary teaching hospital, had compared two increasingly used predictive scores, i.e., neonatal multiple organ dysfunction score (NEOMOD) and the neonatal sequential organ failure assessment (nSOFA) for prediction of mortality and illness severity in neonates with late onset sepsis. Receiver operating characteristic (ROC) curves were plotted at maximum nSOFA and NEOMOD scores attained across all assessment time points, with the area under the curve (AUC) calculated; as well as at specific time points of 24-48 hours before sepsis, at sepsis diagnosis, post 24-48 hours, and at 7-hours after diagnosis.

RESULTS

Of 1851 infants, 419(22.6%) were preterms less than 32 weeks gestation. 88(4.7%) infants had late onset sepsis (LOS), where 81(92%) survived and 7(8%) succumbed. Indices such as base excess, intrauterine growth restriction (IUGR), number of hospital days, presence of periventricular leukomalacia, types of microorganism, platelet count and presence of necrotising enterocolitis were highly associated with mortality. The AUC value for nSOFA's ROC curve was 0.871 with 100% sensitivity and 27.7% specificity when the best cut-off score equalled to '3'. The AUC value for NEOMOD's ROC curve was 0.814 with 85.7% sensitivity and 35.0% specificity when the best cut-off score equaled '4'.

CONCLUSION

Both scoring methods were comparable and prove to be reliable mortality predictive tools. Although the nSOFA score proved to be superior to the NEOMOD score in terms of sensitivity and specificity, the NEOMOD predicted mortality earlier. The NEOMOD was most reliable at 24-48 hours making it a more desirable tool, as nSOFA was most reliable only after 7-days post diagnosis.

Keywords: illness severity score, nSOFA, NEOMOD, neonatal sepsis, mortality

Abstract ID: A-0079

Oral

Mothers' and healthcare professionals' decision-making on breastfeeding choices during the Covid-19 era: A Qualitative Study

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INTRODUCTION

The Covid-19 pandemic has altered the normal course of life, with measures to reduce the virus spread impacting motherhood expectations, and breastfeeding practices.

AIM

This study aimed to explore the perspectives of pregnant mothers and healthcare providers in making the decision for breastfeeding during the pandemic COVID-19.

METHODS

This is a qualitative study via semi-structured in-depth interviews. The study populations were pregnant mothers and healthcare providers of two tertiary referral hospitals in Malaysia. Subjects were selected via purposive sampling to achieve maximal variation. Mothers and healthcare providers fulfilling the inclusion and exclusion criteria were interviewed using a semi-structured topic guide until data saturation was achieved, followed by transcriptions, coding of keywords, and thematic analysis.

RESULTS

Ten pregnant mothers and ten healthcare providers were interviewed either virtually or face-to-face from March 2021 until January 2022. Five main themes, 'perceptions on breastfeeding during pandemic COVID-19', 'guidelines for breastfeeding during pandemic COVID-19', 'importance to protect breastfeeding during pandemic', 'the risk of viral transmission during breastfeeding', and 'involvement of parents in breastfeeding decision during pandemic' emerged from the analysis. The study has shown that breastfeeding practices have been significantly affected since the onset of the pandemic, with confusion regarding available guidelines, especially amongst healthcare providers. All the mothers had expressed their need to give their opinions included in the decision-making process.

CONCLUSION

Although healthcare providers play a critical role in deciding infant feeding choices during the COVID-19 era, shared decision-making between healthcare providers and parents is important as mothers want their opinions to be included in the decision-making process. The study also supports the need to increase public and healthcare providers' awareness of the scientific evidence regarding WHO's COVID-19 breastfeeding recommendations. We recommend the existing local policies and guidelines incorporate the mothers' and healthcare providers' voices and enable shared decision-making.

Keywords: Perspective, Decision-making, Breastfeeding, COVID-19

Abstract ID: A-0080

Oral

The Use of erythropoietin in premature very low birth weight infants amongst seremban population to prevent anaemia of prematurity

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BACKGROUND

Anaemia of prematurity is common in premature infants and the transfusion rate remains high. Recombinant human erythropoietin has been studied in multiple studies and is shown to be safe and effective in reducing the transfusion rate in premature infants.

OBJECTIVE

To assess the effectiveness of recombinant human erythropoietin in reducing the transfusion rate among premature VLBW infants in our neonatal intensive care unit(NICU) after the implementation of subcutaneous erythropoietin.

METHODS

A cross sectional observational study was performed among the premature VLBW infants who were admitted to our NICU from 1stApril 2021 till 1stApril 2022. Demographics, transfusion history and haemoglobin levels were recorded, analysed and compared with the data in 2020 before the introduction of subcutaneous erythropoietin. The dose of erythropoietin administered was according to the Starship clinical guidelines.

RESULTS

There were 137 premature VLBW infants from 1st April 2021 till 1st April 2022, of which 57 infants received erythropoietin according to our study protocol. 6 infants completed 12 doses of erythropoietin, and 32 infants received at least 6 doses. The main reasons for not completing erythropoietin were due to thrombocytosis or discharge once achieved 1.8kg. Among the 137 premature VLBW infants, 74(58%) were transfused. 17(53%) out of the 32 infants who received at least 6 doses of erythropoietin were transfused. In 2020, before erythropoietin was introduced, 74(66%) out of 112 premature VLBW infants were transfused. The mean gestational age and weight for both groups were comparable.

CONCLUSION

Only half of the patients who received at least 6 doses of erythropoietin were transfused compared to 2/3 were transfused before erythropoietin was introduced, although it is not statistically significant. This could be due to the limited sample size of this study and other limitations. Further studies are needed to relook into the efficacy of erythropoietin in reducing transfusion rate among premature infants.

Keywords: erythropoietin. premature very low birth weight infants

Abstract ID: A-0081

E-Poster

Case report : Multisystem inflammatory syndrome in neonates with adrenal insufficiency

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BACKGROUND

Vertical transmission and complications of maternal infections of SARS -COV2 to newborn infants is still under investigation. Most neonates had mild symptoms; however some might present with multisystem organ involvement and shock.

REPORT

We report a case of a male term newborn with Multisystemic Inflammatory Syndrome in Neonate (MIS-N) with adrenal insufficiency.

He was delivered by elective caesarean section, at 38 weeks gestation, for maternal history of 2 previous scars.

The mother had underlying gestational diabetes mellitus with good glycaemic control (HbA1c :5.8%) and no risk of sepsis. She was diagnosed with SAR-COV-2 viral infection at 30 weeks gestation with mild upper respiratory tract symptoms.

The infant had developed symptoms of respiratory distress as early as 10 hours of life, with rapid progression to shock and multi-organ involvement. He showed persistent tachycardia, with cardiogenic shock and early coronary

vessels inflammation and dilatation. He had features of pneumonitis on chest x ray and had required moderate ventilatory setting support.

He also had temperature instability, non specific macular rashes and hypoalbuminaemia. Inflammatory markers were high, with a positive SAR-COV-2 IgG antibody. He was treated with immunoglobulin, steroid and antiplatelet therapy. He was discharged well. The coronary vessels dilatation was transient, showing full recovery by 6 weeks of age. He was on prolonged low dose steroids, with evidence of some adrenal axis suppression, as a part of the complication of treatment.

CONCLUSION

MIS-N could be part of manifestation of SAR-COV-2 infections in neonate. Early index of suspicion with recognition of symptoms will help in diagnosis and management.

Keywords: Multisystem Inflammatory Syndrome in Neonates

Abstract ID: A-0082

Oral

Characteristics and outcomes of newborns with ductal-dependent congenital heart disease in a district hospital with specialist

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BACKGROUND

Neonates with ductal-dependent congenital heart disease (dd-CHD) require a patent ductus arteriosus to maintain adequate circulation and oxygenation. Prostaglandin acts to temporarily maintain the ductus patent until surgical intervention is plausible. As there are limited centres in Malaysia with paediatric cardiology and cardiothoracic surgical expertise, the majority of neonates with dd-CHD are cared for in NICUs without in-house subspecialty support.

OBJECTIVE

To analyse the clinical outcomes and cost implications of managing neonates with dd-CHD prior to cardiac intervention; in a district hospital staffed with paediatricians.

METHODS

All neonates admitted to the NICU at Hospital Sultan Abdul Halim between August 2015 to December 2021; fulfilling the inclusion criteria were included. Demographic data, clinical course and prostaglandin therapy details were retrospectively obtained from electronic medical records.

RESULTS

Fifty-three patients with dd-CHD receiving prostaglandin therapy were included with a male-to-female ratio of 1.12:1. The most common diagnoses were pulmonary atresia/stenosis (n=16, 30.2%), transposition of great arteries with intact ventricular septum (n=14, 26.4%) and aortic arch anomalies (n=9, 17%). Forty-one (77.4%) survived after surgery with the current mean age of 3.3 years old (+/-2.0). From the study population, 28 have undergone palliative surgery (52.8%), 14 received corrective surgery (26.4%) and 5 received conservative care (9.4%) while 6 succumbed prior to surgery (11.3%). Seventy-percent received their first cardiac intervention after day 10 of life. We found a positive correlation between an increased prostaglandin usage and number of long lines requirement with a longer waiting time to surgery ($p = 0.01$). Cost-analysis reveals an expenditure of approximately RM3 million on prostaglandin alone within the study period.

CONCLUSION

Neonates with dd-CHD receiving prostaglandin in a district hospital while awaiting surgical intervention generally have good outcomes. However, this is associated with high cost expenditure. Expansion of regional paediatric cardiology and cardiothoracic services may reduce cost, waiting times, morbidity and mortality rates.

Keywords: Congenital heart defect (CHD), Patent ductus arteriosus (PDA), Prostaglandin, Newborns, Neonates

Abstract ID: A-0083

E-Poster

Pregnancy permission criteria and perinatal outcomes in women with systemic lupus erythematosus

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BACKGROUND

Autoimmune diseases predominantly affect women and often develop in women of reproductive age. For patients with systemic lupus erythematosus (SLE), there is no fixed criteria for the disease remission status and remission duration concerning pregnancy permission. Therefore, patients must be handled on a case-by-case basis, considering their risks.

OBJECTIVE

In this study, we investigated the pregnancy permission criteria based on perinatal outcomes of pregnancies complicated by SLE at our hospital.

METHODS

We retrospectively reviewed 20 women who had pregnancies complicated by SLE and delivered at our hospital in the past 5 years. Differences were analyzed using a one-way Analysis of Variance (ANOVA) followed by the Bonferroni-Dunn test for multiple comparisons. $P < 0.05$ was considered significant.

RESULTS

Upon comparing a group with perinatal complications (obstetric complications and fetus and fetal appendages abnormalities) (10 patients) and a group without them (10 patients), no significant difference was observed in the maternal age, height, weight (non-pregnant and pre-partum), body mass index (non-pregnant and pre-partum), presence and absence of smoking. The pre-pregnancy estimated glomerular filtration rate (eGFR) and early pregnancy urinary protein levels were significantly higher ($P < 0.05$). No significant difference between the two groups was observed in the comorbid antiphospholipid syndrome, lupus nephritis, and the number of patients with anti-SS-A antibodies. However, patients who did not meet the pregnancy permission criteria were significantly more in the group with perinatal complications ($P < 0.05$). Four patients who did not meet the pregnancy permission criteria became pregnant, and all fetuses had poor outcomes (two intrauterine fetal deaths, one neonatal death, and one infant death).

CONCLUSION

Our findings suggested that pre-pregnancy eGFR and early pregnancy proteinuria were important predictors of perinatal outcome.

Keywords: pregnancy in SLE, perinatal outcomes

Abstract ID: A-0084

Oral

Changes in fetal lung size and survival in fetuses with fetoscopic endotracheal occlusion for severe congenital diaphragmatic Hernia

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OBJECTIVE

To describe the correlation between survival of children with severe left-sided congenital diaphragmatic hernia (CDH) and the observed/expected lung-to-head ratio (o/e LHR) measured by ultrasound before versus after fetoscopic endotracheal occlusion (FETO).

METHODS

Children with severe left-sided CDH (o/e LHR <25%) who underwent FETO in our hospital in 2014–2022 were included. Fetuses with chromosomal abnormalities, who died, and in whom balloon occlusion failed were excluded. FETO was performed at 28–29 weeks' gestation in severe left-sided CDH. The primary outcome was survival at 6 months after birth. The o/e LHR before FETO and before balloon removal were collected retrospectively, are shown as median (minimum to maximum), and were compared using the Mann-Whitney U test.

RESULTS

Fifteen children were included; of them, five (33%) survived. The median gestational age at delivery was 35⁺⁵ (31⁺⁶ to 38⁺⁵) weeks. Median o/e LHR values before FETO and before balloon removal were 20% (8–24%) and 25% (14–62%), respectively. The o/e LHR before FETO in survivors and non-survivors were 21% (20–24%) and 20% (8–23%) ($p=0.04$), respectively. The o/e LHR before balloon removal in survivors and non-survivors were 48% (37–62%) and 24% (14–34%) ($p<0.01$), respectively. Survival rates were 0% and 50% in children with o/e LHR of <20% and >20% before FETO, respectively. Survival rates were 0% and 100% among children with o/e LHR of <35% and >35% before balloon removal, respectively.

CONCLUSION

Fetal lung size was greater in survivors versus non-survivors. FETO increased fetal lung size and survival rates of some children with severe left-sided CDH.

Keywords: congenital diaphragmatic hernia, fetoscopic endotracheal occlusion

Abstract ID: A-0085

E-Poster

Gluteal monster - A case report

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BACKGROUND

This case is presented because of its low incidence and its rarity. Although infrequent, sacrococcygeal teratoma is the most common germ cell malignancy in newborns with a prevalence of 1 in 40000, with female preponderance (4:1), arising from the caudal end of spine.

CASE REPORT

This newborn is the fourth child of non-consanguineous parents. She was born with a good Apgar score of 9/10 via vaginal delivery at 39 weeks with a healthy weight of 4150g. She appeared comfortable at rest, not dysmorphic, pink, with grossly evident abnormal mass over bilateral gluteal region joined in the midline measuring 15cm in length and 12cm across. On examination, the spine and the female genitalia were normal, and she had good urine and bowel output. Other systemic examinations were unremarkable.

Her mother is a 40 year old, para 4 lady who had an uneventful pregnancy. Urgent ultrasound scan of the spine and pelvis revealed a solid cystic complex sacral/presacral mass with no direct extension of the mass into the spinal canal seen. Serum alpha fetoprotein (AFP) was >2479 IU/ml. She was immediately transferred to the Pediatric Surgical team and an operation was done at day 5 of life. Specimens sent for histopathological examination confirmed the diagnosis of sacrococcygeal mature cystic teratoma. The AFP level has dropped to 300IU/ml at day 38 of life.

CONCLUSION

The main treatment for sacrococcygeal teratoma, regardless of histological type, is a complete resection of the tumor and the coccyx; risk of recurrence is extremely high if not performed. Sometimes a mature or immature teratoma also contains malignant cells. It is important to recognize the existence of this pathology in order to have the clinical expertise that offers timely diagnosis, an appropriate and multidisciplinary treatment. Monitoring with alpha-fetoprotein and ultrasound is a key to detect recurrence or postoperative complications

Keywords: sacrococcygeal, teratoma

Abstract ID: A-0087

E-Poster

A type III CPAM in the House- Is it a smooth sailing journey?

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BACKGROUND

Congenital pulmonary airway malformation (CPAM) type III is one out five variant and it is the second rarest type after type 0 with the prevalence ranging from 5-10%. The aetiology is unknown.

CASE REPORT

A 32-year old Malay lady, primigravida was diagnosed antenatally to have right CPAM at 29 weeks. A female baby was born at 36 weeks. The delivery was induced due to the mother having severe pre-eclampsia (PE) with IUGR. The baby was admitted to NICU for multiple problems. The primary diagnosis was right CPAM. High resolution computed tomography (HRCT) of thorax revealed the diagnosis and proceed with right thoracotomy and lobectomy. The histology confirmed lung tissue with multicystic lesion diagnostic of type 3 CPAM. She was doing well and was discharged at 3 months old with home oxygen therapy.

CONCLUSION

CPAM is a rare congenital lung disorder. Surgical excision is recommended to make a definite diagnosis, and histopathological examination is crucial to differentiate the classification of the lesion.

Keywords: Congenital pulmonary airway malformation, CPAM

Abstract ID: A-0088

E-Poster

Successful versus failed instrumental delivery; Predictors and obstetric outcomes: A single-center experience

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BACKGROUND

Instrumental delivery either with forceps or vacuum may facilitate vaginal birth and reduce complications associated with caesarean delivery. However, failed instrumental delivery followed by caesarean section can be associated with a significantly higher rate of maternal and fetal morbidities, if compared to spontaneous delivery or successful instrumental delivery. For this study, the procedure was considered failed when the delivery could not be achieved with the initial instrument and a different vacuum cup, forceps or caesarean delivery was needed.

OBJECTIVES

The objective of this study was to identify the predictors of successful or failed instrumental delivery and to evaluate the maternal and fetal outcomes in both groups.

METHODS

This was a two-year retrospective study at Kemaman Hospital in Terengganu, Malaysia (2020 and 2021). There were a total of 8068 deliveries where 219 (2.7%) underwent instrumental delivery. All records of patients who had instrumental delivery were reviewed and data were obtained from the medical records. Maternal age, parity, gestational age at delivery, vaginal examination

findings, maternal indication for instrumental delivery, intrapartum use of ultrasound, type of forceps or vacuum used and seniority of the obstetricians were all documented. Instrumental delivery records were analyzed based on whether the instrumentation was successful or failed. The maternal and foetal outcomes were also recorded.

RESULTS

Maternal age, parity, gestational age at delivery, occiput posterior position and foetal distress were not significantly different in both groups ($p > 0.005$). With the use of ultrasound during instrumental delivery, the number of successful was significantly higher than failed instrumental delivery ($p = 0.003$). In both groups there was no significant difference in maternal and neonatal outcomes.

CONCLUSION

The use of ultrasound prior to the application or consideration of instruments was the most important factor in the outcome of instrumental delivery. The difference in maternal or fetal outcomes were not statistically significant in either group.

Keywords: instrumental delivery, predictors, obstetrics outcome

Abstract ID: A-0089

Oral

A cross-sectional study on the maternal outcome of an emergency caesarean section at different time intervals in a tertiary hospital

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BACKGROUND

Caesarean section (CS) is more likely to result in adverse maternal outcomes (AMO) than vaginal delivery, especially if performed in an emergency manner. The question is whether providing emergency CS (emCS) after office hours increases the risk of AMO.

OBJECTIVE

To investigate the relationship between the timing of emCS delivery was performed and AMO.

METHODS

This cross-sectional study included 211 low-risk mothers who had their babies delivered via emLSCS at Hospital Tunku Azizah Kuala Lumpur between August 1st and August 31st 2022. Those delivered via elective or semi-emergency CS were excluded. All data were gathered from the daily operating theatre (OT) census and the hospital information system (HIS). They were divided into three groups based on the time the patient arrived at the OT. The hours

of operation were 0700-1601 during the day, 1601H-0000H at night, and 0001H-0659H in the early morning. PPH, viscus injury, sepsis within 48 hours of delivery, unplanned ICU admission, relaparotomy, and hysterectomy were all recorded and analysed using SPSS.

RESULTS

The prevalence of composite AMO among emCS obtained was 5.20%, which was consistent with previously reported studies where emCS complication was known to be very low, particularly in low-risk mothers. PPH was the most common, accounting for 11 cases and one ICU admission following a caesarean hysterectomy. There had been no reports of viscus injury or sepsis within 48 hours of delivery. There was no statistically significant correlation between delivery via emCS after office hours and composite AMO. The adjusted odds ratios for emCS in the evening and early morning were 1.12 (95 percent CI 0.25-5.04) and 0.75 (95 percent CI 0.14-3.96), respectively.

CONCLUSION

There was no significant correlation between adverse maternal outcomes and the time of emergency caesarean section in low-risk mothers.

Keywords: emergency caesarean, off-hours, working hours, adverse maternal outcomes, complications

Abstract ID: A-0091

E-Poster

Case report series on successful antibiotic lock therapy for catheter salvage and treatment of central line associated bloodstream infection

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Antibiotic Lock Therapy (ALT) is indicated for patients with catheter related bloodstream infections involving long term catheters usage where catheter salvage is the goal. A highly prepared concentrated antibiotic is 'locked' to sterilize a previously infected catheter in order to penetrate or disrupt a biofilm in the catheter lumen. Whilst some of the antibiotics have been extensively studied in lock solutions, the usage of piperacillin/tazobactam and carbapenem is still lacking. We described our experience of using these 2 antibiotics as ALT in our chronic patients for catheter salvage and treatment of central line associated bloodstream infection (CLABSI).

Case 1 involved a late-preterm baby who was treated for MAS with PPHN from birth. Echo confirmed a diagnosis of tetralogy of fallot (TOF) with infundibular pulmonary stenosis and he was started on intravenous prostaglandins (PGE2) from day 1 of life. Due to PGE2 dependence, he was on multiple central venous catheters that were changed every 2 weeks. He required prolonged antibiotics for treatment of subacute infective endocarditis likely related to CLABSI. Despite more than 4 weeks therapy the blood cultures still grew persistent *Elizabeth miricola*. He was embarked on piperacillin/tazobactam ALT for 2 weeks. The ALT proved successful and he was able to be tapered off PGE2 at 6 months of age.

Case 2 involved a term baby with confirmed DiGeorge Syndrome and T cell defect with TOF and pulmonary atresia who required PGE2 since day 1 of life. She suffered from multiple nosocomial septicaemia and myositis. Subsequent series of blood cultures were positive for organisms including *Bulkhoderia cepacia*, *Acinetobacter baumannii* MDR and *Serratia marcerens*. Due to difficult intravenous access, she was embarked on meropenem ALT for catheter salvage which turned out successful. She was able to be tapered off the PGE2 at 5 months of age.

Keywords: Central Line Associated Bloodstream Infection (CLABSI), Antibiotic lock therapy (ALT), Central venous catheter (CVC), piperacillin/tazobactam, meropenem

Abstract ID: A-0092

E-Poster

Oropharyngeal teratoma in a newborn: A rare disorder

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BACKGROUND

Teratoma is a rare neoplasm derived from the germinal layers of the embryo. There are few common sites for teratomas identified, however, oropharyngeal teratoma is the rarest site (2% of all teratomas). Teratoma develops when there is disruption during the cell differentiation process.

CASE REPORT

We present a neonatal case of oropharyngeal teratoma. A baby boy 2800g, was born prematurely at 35weeks gestation to a 38-year-old mother via emergency caesarean section for acute fetal distress. The mother has underlying gestational diabetes mellitus on metformin and s/c insulatard. Prenatal findings did not identify the tumours except polyhydramnios. The newborn exhibited respiratory compromise immediately after birth. However due to the distorted anatomy of the oropharynx on direct visualization that presented as small masses protruding from the oropharyngeal area, he underwent multiple attempts of intubation before being successfully intubated under controlled environment in the neonatal intensive care unit. Urgent computed tomography of the neck (CT) showed an ill-defined hypodense non-enhancing cystic lesion mass (2.0x1.6x1.4cm) at the left posterior pharyngeal wall extending to

the prevertebral spaces (0.9x0.6x0.9cm) with narrowing of the oropharynx. He underwent complete resection of the pharyngeal mass during the first week of life. Histologically the oropharyngeal mass was identified as mature cystic teratoma. The age-adjusted alpha-fetoprotein level was normal. He was successfully extubated on day 23 of life. Following the surgery, he required supplemental oxygen and successfully weaned from oral gastric tube feeding to oral feeding.

CONCLUSION

Oropharyngeal teratomas have benign histopathology and may cause airway obstruction, respiratory compromise and feeding or swallowing difficulties. Mature teratoma is usually noncancerous and has a possibility of recurrence once surgically removed. Long-term follow-up is warranted to monitor complications or recurrence of the disease.

Keywords: oropharyngeal, neonate, respiratory compromise, mature teratoma

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